

Therapy for Stage IV Non–Small Cell Lung Cancer Without Driver Alterations: ASCO Living Guideline, 2026.3.0

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ABSTRACT

ASCO Guidelines provide recommendations with comprehensive review and analyses of the relevant literature for each recommendation, following the guideline development process as outlined in the *ASCO Guidelines Methodology Manual*. ASCO Guidelines follow the *ASCO Conflict of Interest Policy for Clinical Practice Guidelines*.

Clinical Practice Guidelines and other guidance (“Guidance”) provided by ASCO is not a comprehensive or definitive guide to treatment options. It is intended for voluntary use by clinicians and should be used in conjunction with independent professional judgment. Guidance may not be applicable to all patients, interventions, diseases, or stages of diseases. Guidance is based on review and analysis of relevant literature and is not intended as a statement of the standard of care. ASCO does not endorse third-party drugs, devices, services, or therapies and assumes no responsibility for any harm arising from or related to the use of this information. See complete disclaimer in *Appendix 1* and *Appendix 2* (online only) for more.

PURPOSE To provide evidence-based recommendations for patients with stage IV non–small cell lung cancer (NSCLC) without driver alterations.

METHODS This ASCO living guideline offers continually updated recommendations based on an ongoing systematic review of randomized controlled trials (RCTs), with the latest time frame spanning March–October 2025. An Expert Panel of medical oncology, pulmonary, community oncology, research methodology, and advocacy experts was convened. The literature search included systematic reviews, meta-analyses, and RCTs. Outcomes of interest include efficacy and safety. Expert Panel members used available evidence and informal consensus to develop evidence-based guideline recommendations.

RESULTS This guideline consolidates all previous updates and reflects the body of evidence informing this guideline topic. Six new RCTs were identified in the latest search of the literature to date.

RECOMMENDATIONS Evidence-based recommendations were updated to address first, second, and subsequent treatment options for patients without driver alterations. Additional information is available at www.asco.org/thoracic-cancer-guidelines.

ACCOMPANYING CONTENT

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 Appendix

 Data Supplement

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INTRODUCTION

The treatment of patients with advanced stage IV non–small cell lung cancer (NSCLC) has evolved dramatically in the past decade. ASCO launched living practice guidelines for systemic management for patients with stage IV NSCLC with¹

and without driver alterations² in 2022, both of which have been updated recently to reflect the rapidly evolving treatment landscape.^{3–17} The purpose of this guideline is to update the living guidelines on the systemic management of patients with stage IV NSCLC without driver alterations. This comprehensive update is based on evidence with potential impact

TARGET POPULATION AND AUDIENCE

Target Population

Patients with stage IV NSCLC without targetable driver alterations.

Target Audience

Oncology care clinicians (including primary care physicians, specialists, nurses, social workers, and any other relevant member of a comprehensive multidisciplinary cancer care team), patients, and their caregivers.

- Interventions: Chemotherapy, monoclonal antibodies, immunotherapy, palliative care, and no treatment.
- Comparisons: Chemotherapy, monoclonal antibodies, immunotherapy, palliative care, and no treatment.
- Outcomes: Included progression-free survival (PFS), overall survival (OS), treatment toxicity (adverse events [AEs]; usually grade 3 to 4 AEs), objective response rate (ORR), and quality of life (if reported).
- Study design: Systematic reviews, phase II or III RCTs, single-arm phase II studies.
- Sample size: Minimum sample size of 20 patients for immune checkpoint therapy and 50 patients for chemotherapy.

Articles were excluded from the systematic review if they were (1) meeting abstracts not subsequently published in peer-reviewed journals; (2) editorials, commentaries, letters, news articles, case reports, and narrative reviews; and (3) published in a non-English language.

One full panel meeting and three subgroup panel meetings were held, and members were asked to provide ongoing input on the guideline development protocol, quality and assessment of the evidence, generation of recommendations, draft content, as well as review and approve drafts during the entire development of the guideline. ASCO staff met routinely with the Expert Panel co-chairs and corresponded with the Panel via e-mail to coordinate the process to completion. Ratings for the strength of the recommendation and evidence quality are provided with each recommendation, defined in Appendix [Table A2](#). The quality of the evidence for each outcome was assessed using the Cochrane Risk of Bias tool and elements of the GRADE quality assessment and recommendations development process.^{18,19} GRADE quality assessment labels (ie, high, moderate, low, very low) were assigned for each outcome by the project methodologist in collaboration with the Expert Panel co-chairs and reviewed by the full Expert Panel. All funding for the administration of the project was provided by ASCO.

Guideline Review and Approval

The draft recommendations were released to the public for open comment from August 21 through September 3, 2025. Response categories of “Agree as written,” “Agree with suggested modifications,” and “Disagree. See comments” were captured for every proposed recommendation with nine respondents and no written comments received. Of the 47 recommendations included in the open comment, there were no respondents who disagreed with a recommendation; the remaining 100% either agreed or agreed with slight modifications to the recommendations. Expert Panel members reviewed comments from all sources and determined whether to maintain original draft recommendations, revise with minor language changes, or consider major recommendation revisions.

on practice published since the last update by Jaiyesimi et al,⁶ and subsequent rapid updates published after.^{7,8}

Our ability to generalize clinical trial results to real-world populations, including both efficacy and safety, relies on broad enrollment of patients to clinical trials. These guidelines are intended for everyday practice, acknowledging that patients may have special circumstances, which are not included in the published evidence base.

GUIDELINE QUESTIONS

This clinical practice guideline update addresses two overarching clinical questions: (1) What are the most effective first-line treatment options for patients with stage IV non-small cell lung cancer (NSCLC) without driver alterations, based on cancer subtype? (2) What are the most effective second-line and subsequent treatment options for patients with stage IV NSCLC without driver alterations based on cancer subtype?

METHODS

Guideline Development Process

This systematic review-based guideline product was developed by a multidisciplinary Expert Panel, which included a representative from Ontario Health (Cancer Care Ontario), a patient representative, and an ASCO guidelines staff member with health research methodology expertise (Appendix [Table A1](#), online only).

The recommendations were developed by using a systematic review of evidence identified through online searches of PubMed (the latest being March 2025–October 2025) of phase III randomized controlled trials (RCTs). Articles were selected for inclusion in the systematic review based on the following criteria:

- Population: Patients with stage IV NSCLC whose test results show:
 - a. PD-L1 tumor proportion score (TPS) test results available to the clinician being optimal and without driver alterations.

All changes were incorporated into the final manuscript before ASCO Evidence Based Medicine Committee (EBMC) review and approval. All ASCO guidelines are ultimately reviewed and approved by the Expert Panel and the ASCO EBMC before submission to the *Journal of Clinical Oncology* for editorial review and consideration for publication.

Living Guideline Updating

The ASCO Expert Panel and guidelines staff will work with co-chairs to conduct a formal review of the emerging literature every 2 weeks to assess the necessity and scope of update. The ASCO Guidelines Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the living guideline update process. This is the most up-to-date information as of the publication date.

Versioning and Numbering Statement

To ensure transparency and facilitate the tracking of evidence updates, ASCO has implemented a structured numbering system for living guidelines. This system reflects the status of the recommendations as follows:

1. Major Updates (eg, 2026.2.0): These represent full updates or significant revisions approved by the Expert Panel and the Evidence Based Medicine Committee (EBMC). They typically incorporate comprehensive literature reviews and involves resubmission of the entire guideline.
2. Rapid Updates (eg, 2026.2.1): These “mini updates” serve as timely notifications of revised recommendations or supporting evidence, ensuring the guideline remains current between full update cycles.

RESULTS

Characteristics of Studies Identified in the Literature Search

Of the 73 studies identified in the latest literature search that spanned from March to September 2025, six RCTs met the eligibility criteria for this update. The randomized trials compared different therapies for nondriver alterations.^{20–25} Results from these studies either necessitated some modifications to the recommendations, or the quality of evidence rating in some recommendations. Discussions of evidence supporting unchanged recommendations have been reviewed and updated from previous guideline publications.^{3–8}

RECOMMENDATIONS

Based on the body of evidence identified to date through several update iterations, complete recommendations informing the treatment of patients with stage IV NSCLC without driver alterations are offered in [Table 1](#).

BIOMARKER TESTING

Recommendation 1.0

The literature review and clinical interpretation supporting this recommendation are detailed in the driver alteration version of this guideline.²⁶

FIRST-LINE THERAPY

Nonsquamous Cell Carcinoma

PD-L1 Expression, TPS ≥50%

Recommendation 1.1.

Literature review update. Three open-label, phase III randomized trials have evaluated PD-(L)1 monotherapy versus chemotherapy in patients with tumor expression of PD-L1 ≥50%. These include the IMpower110 (N = 205),^{27,28} KEYNOTE-024 (N = 305),^{29,30} and EMPOWER-Lung 1 (N = 563)^{22,31,32} studies using atezolizumab, pembrolizumab, and cemiplimab, respectively. The results from these studies showed consistent significant improvement in both PFS and OS for patients who received immune checkpoint inhibitors (ICIs) compared to chemotherapy with IMpower110 reporting median OS of 20.2 months (95% CI, 17.2 to 27.9) for atezolizumab in the PD-L1 high subset; KEYNOTE-024 reporting 26.3 months (95% CI, 18.3 to 40.4) for pembrolizumab; and EMPOWER-Lung 1 reporting 26.1 months (95% CI, 22.1 to 31.9) for cemiplimab. Both the EMPOWER-Lung 1 and KEYNOTE-024 studies have reported 5-year OS updates, demonstrating a 5-year OS of 29.0% for cemiplimab and 31.9% for pembrolizumab.

Clinical interpretation. The proportion of patients with nonsquamous NSCLC treated with ICI in each study varied: 81.2% in KEYNOTE-024, 74.8% in IMpower110, and 57% in EMPOWER-Lung 1. Irrespective of the different assays employed and minor baseline patient characteristic differences, the OS benefit of single-agent PD-(L)1 inhibitor over chemotherapy was clearly demonstrated across the three studies. An active area of ongoing research includes analysis of biomarkers for patients within these cohorts who could benefit from escalation of treatment, including mutations in *STK11/KEAP1*, although currently these are limited to retrospective analyses.³³ In the absence of head-to-head trials, it is not clear which patients within this cohort would benefit from the addition of chemotherapy to ICI monotherapy, and given the increased risks associated with chemotherapy, the recommendation is to offer PD-(L)1 monotherapy to patients with high expression of PD-L1. A phase III study randomly assigned patients with metastatic NSCLC with PD-L1 TPS >50% to ipilimumab and pembrolizumab versus pembrolizumab-placebo and found no difference in OS with the combination (21.4 months v 21.9 months; hazard ratio [HR], 1.08 [95% CI, 0.85 to 1.37]; *P* = .74), but did find

increased risk of AEs with dual checkpoint inhibition.³⁴ An additional area of investigation is whether ivonescimab, a vascular endothelial growth factor (VEGF) PD-1 bispecific antibody, is superior to anti-PD-1 monotherapy in the PD-L1-high population. Ivonescimab is approved in China as monotherapy in PD-L1-high NSCLC based on results of the HARMONi-2 study,³⁵ which demonstrated superior PFS with trend favoring OS for ivonescimab compared to pembrolizumab. Global studies in NSCLC are underway in various patient populations.

Recommendations 1.2 and 1.3.

Literature review update. In EMPOWER-Lung 3, 904 patients with advanced or metastatic, epidermal growth factor receptor (EGFR) and anaplastic lymphoma kinase (ALK) mutation-negative NSCLC were randomly assigned in a 2:1 fashion to platinum-doublet chemotherapy plus cemiplimab 350 mg once every 3 weeks versus chemotherapy plus placebo followed by cemiplimab versus placebo maintenance (pemetrexed maintenance was allowed for the nonsquamous histology). 57.1% of patients on this study had nonsquamous histology. The study met its preset criteria for OS, with a median OS of 21.1 months with cemiplimab compared to 12.9 months with chemotherapy (HR, 0.65 [95% CI, 0.51 to 0.82]). Among the PD-L1-high subset (irrespective of disease histology), the OS was 23.5 months (HR, 0.61 [95% CI, 0.37 to 1.02]), favoring cemiplimab + chemotherapy over placebo + chemotherapy. At the 2-year follow-up update, this benefit was maintained with an OS HR of 0.56 (95% CI, 0.36 to 0.86).³⁶ The KEYNOTE-021, Cohort G, and KEYNOTE-189 studies both only enrolled patients with advanced or metastatic nonsquamous NSCLC treated with carboplatin plus pemetrexed and pembrolizumab in the experimental group. At the 5-year update from KEYNOTE-189, in the PD-L1-high subset, median OS was 27.7 months for pembrolizumab plus chemotherapy compared to 10.1 months with placebo plus chemotherapy (HR, 0.68 [95% CI, 0.49 to 0.96]), with a 5-year landmark OS of 27.7% in the pembrolizumab arm.³⁷

Clinical interpretation. KEYNOTE-189 was a phase III placebo-controlled study and patients were randomly assigned to platinum (cisplatin or carboplatin)-pemetrexed-pembrolizumab versus platinum-pemetrexed-placebo followed by pemetrexed-pembrolizumab versus pemetrexed-placebo maintenance. In EMPOWER-Lung 3, all NSCLC histologies were enrolled. Furthermore, all studies enrolled patients irrespective of PD-L1 expression (which were both assessed by the Dako 22C3 assay), and therefore, OS based on PD-L1 expression represents a subset analysis. All three studies demonstrated a significant improvement of OS with a PD-1 inhibitor. However, the studies do not shed clarity as to when platinum-pemetrexed should be added to pembrolizumab or cemiplimab in the setting of high PD-L1 expression. Expert consensus is to consider cytotoxic chemotherapy if there is significant tumor burden or risk for visceral crisis with delayed response to therapy, although these treatment strategies have not been compared head-

to-head in a clinical trial. The Eastern Cooperative Oncology Group (ECOG)-American College of Radiology Imaging Network (ACRIN) INSIGNA trial, which recently completed accrual, will address the question of which subsets of patients, if any, benefit from the addition of chemotherapy to PD-(L)1 therapy.

Recommendation 1.4.

Literature review update. The IMpower130 and IMpower150 studies evaluated the benefit of the addition of the PD-L1 inhibitor atezolizumab to platinum-doublet chemotherapy with or without bevacizumab in non-squamous advanced or metastatic NSCLC. In IMpower130,³⁸ 724 patients were randomly assigned in a 2:1 fashion to receive 4-6 cycles of carboplatin, nab-paclitaxel, and atezolizumab versus chemotherapy alone followed by atezolizumab or best supportive care versus switch pemetrexed maintenance (at the investigator's discretion). Crossover to atezolizumab was allowed for the patients in the chemotherapy group before a prespecified date cutoff. In the PD-L1-high subset, the median OS was 17.3 months in the atezolizumab-containing group, compared to 16.9 months in the chemotherapy group (HR, 0.84 [95% CI, 0.51 to 1.39]). In the IMpower150 study,^{39,40} 1,202 patients with advanced or metastatic nonsquamous NSCLC were randomly assigned 1:1 to receive carboplatin-paclitaxel-atezolizumab (ACP), ACP + bevacizumab (ABCP), or carboplatin-paclitaxel-bevacizumab (BCP). Patients received 4-6 cycles of induction with chemotherapy followed by atezolizumab, or bevacizumab, or both in maintenance until disease progression. Crossover to atezolizumab was not allowed. Patients with PD-L1-high (tumor cells [TC] 3 or immune cells [IC] 3) had longer OS with an atezolizumab-containing regimen (ACP median OS, 26.3 months; ABCP median OS, 30.0 months) compared to BCP (15.0 months). The IMpower132 trial,⁴¹ which randomly assigned 578 patients 1:1 to platinum-pemetrexed-atezolizumab (APP) followed by pemetrexed-atezolizumab maintenance or platinum-pemetrexed followed by pemetrexed maintenance, was negative, and is not currently US Food and Drug Administration (FDA) approved.

Clinical interpretation. Understanding the relative contribution to OS of bevacizumab and atezolizumab in the IMpower150 regimen is difficult to ascertain. In the final OS analysis, there was no statistically significant difference between chemotherapy and atezolizumab versus chemotherapy and bevacizumab. The improvement in OS is seen with chemotherapy plus atezolizumab and bevacizumab but there is no comparison of ABCP to ACP, only of ABCP to BCP. Furthermore, the addition of bevacizumab does increase risk—there were 5 deaths in the ABCP group attributed to pulmonary hemorrhage or hemoptysis, a known significant adverse effect of bevacizumab. To date, there have not been any studies that have addressed synergy of VEGF and PD-L1 inhibition specifically in the PD-L1-high subgroup. However, combination VEGF and PD-(L)1 inhibition is an area of active research and clinical

TABLE 1. Summary of All Recommendations

Category	Recommendation
<i>General Note.</i> The following recommendations (strong or conditional/weak) and terminology (Data Supplement) represent reasonable options for patients depending on clinical circumstances and in the context of individual patient preferences. Recommended care should be accessible to patients whenever possible. For recommendations with multiple treatment options of the same evidence quality and strength of recommendation, the decision of which agent to offer should be tailored based on discussion of efficacy, toxicity, drug access, and cost with each patient. Unless otherwise specified the recommended number of cycles of platinum doublet chemotherapy is four.	
What are the most effective first-line treatment options for patients with good performance status and the following biomarkers?	
Biomarker testing	1.0. Biomarker testing with a validated tissue and/or blood-based broad multigene panel and a validated tissue IHC assay for PD-L1, HER2, and MET protein expression should be universally accessible for all patients diagnosed with NSCLC. (Evidence quality: High; Strength of recommendation: Strong) <i>Qualifying Statement for Recommendation 1.0:</i> Combination of blood and tissue testing may maximize chance of detecting molecular alterations. Comprehensive RNA-based NGS testing ideally incorporating RNA-based NGS for detecting genomic alterations is preferred. Tissue testing has the advantage of histologic assessment and should be attempted when feasible, and the false-negative rate of liquid biopsy should be considered. PD-L1 IHC alone should not be used to guide treatment decisions.
Nonsquamous cell carcinoma	
PD-L1 expression TPS ≥50%	<i>Clinicians should offer any of the following single-agent immune checkpoint inhibitors:</i> 1.1. Pembrolizumab or cemiplimab or atezolizumab. (Evidence quality: High; Strength of recommendation: Strong) <i>Other combination regimens that may be offered in select instances include:</i> 1.2. Pembrolizumab + carboplatin + pemetrexed. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.3. Cemiplimab + carboplatin + pemetrexed. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.4. Atezolizumab + carboplatin + (nab)-paclitaxel ± bevacizumab (in the absence of contraindications to bevacizumab). (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.5. Nivolumab and ipilimumab. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.6. Nivolumab and ipilimumab with two cycles of platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.7. Durvalumab and tremelimumab plus platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
PD-L1 expression, TPS 1%-49%	<i>Clinicians should offer any of the following combination immune checkpoint inhibitor and chemotherapy regimens:</i> 1.8. Pembrolizumab + carboplatin + pemetrexed. (Evidence quality: Moderate; Strength of recommendation: Strong) 1.9. Cemiplimab + carboplatin + pemetrexed. (Evidence quality: Moderate; Strength of recommendation: Strong) <i>Other combination regimens that may be offered in select instances include:</i> 1.10. Atezolizumab + carboplatin + (nab)-paclitaxel ± bevacizumab (in the absence of contraindications to bevacizumab). (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.11. Nivolumab and ipilimumab. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.12. Nivolumab and ipilimumab plus two cycles of platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.13. Durvalumab and tremelimumab plus platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.14. Pembrolizumab monotherapy may be offered to patients who are ineligible for or decline combination therapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
Unknown or negative PD-L1 expression, TPS <1%	<i>Clinicians may offer any of the following:</i> 1.15. Pembrolizumab + carboplatin + pemetrexed. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.16. Cemiplimab + carboplatin + pemetrexed. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.17. Atezolizumab + carboplatin + (nab)-paclitaxel ± bevacizumab (in the absence of contraindications to bevacizumab). (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.18. Nivolumab and ipilimumab plus two cycles of platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.19. Nivolumab and ipilimumab. (Evidence quality: Moderate; Strength of recommendation: Conditional) 1.20. Durvalumab and tremelimumab plus platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)

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TABLE 1. Summary of All Recommendations (continued)

Category	Recommendation
Squamous cell carcinoma	
PD-L1 expression, TPS ≥50%	<i>Clinicians should offer treatment with single-agent immune checkpoint inhibitors:</i>
	2.1. Single-agent pembrolizumab or cemiplimab or atezolizumab. (Evidence quality: High; Strength of recommendation: Strong)
	<i>Other combination regimens that may be offered in select instances include:</i>
	2.2. Pembrolizumab + carboplatin + paclitaxel (or nab-paclitaxel) or cemiplimab + carboplatin + paclitaxel. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.3. Nivolumab and ipilimumab. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.4. Nivolumab and ipilimumab plus two cycles of platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
PD-L1 expression, TPS 1%-49%	2.5. Durvalumab and tremelimumab plus platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	<i>Clinicians should offer any of the following combination immune checkpoint inhibitor and chemotherapy regimens:</i>
	2.6. Pembrolizumab + carboplatin + paclitaxel (or nab-paclitaxel). (Evidence quality: Moderate; Strength of recommendation: Strong)
	2.7. Cemiplimab + carboplatin + paclitaxel. (Evidence quality: Moderate; Strength of recommendation: Strong)
	2.8. Nivolumab and ipilimumab. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.9. Nivolumab and ipilimumab plus two cycles of platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
Unknown or negative PD-L1 expression, TPS <1%	2.10. Durvalumab and tremelimumab plus platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.11. Pembrolizumab monotherapy may be offered to patients who are ineligible for or decline combination therapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	<i>Clinicians may offer any of the following:</i>
	2.12. Pembrolizumab + carboplatin + paclitaxel (or nab-paclitaxel). (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.13. Cemiplimab + carboplatin + paclitaxel. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.14. Nivolumab and ipilimumab plus two cycles of platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
General approaches	2.15. Nivolumab and ipilimumab. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.16. Durvalumab and tremelimumab plus platinum-based chemotherapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	3.1. Patients with advanced lung cancer should be referred to interdisciplinary palliative care teams (consultation) that provide inpatient and outpatient care early in the course of disease, alongside active treatment of their cancer. (Evidence quality: High; Strength of recommendation: Strong)
Patients with contraindications to bevacizumab	3.2. For patients who are not candidates for immune checkpoint inhibitor therapy, clinicians should offer platinum doublet combination therapy for patients with preserved PS. (Evidence quality: High; Strength of recommendation: Strong)
	3.3. Clinicians may offer nonplatinum therapy combinations for patients who have contraindications to platinum therapy. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	3.4. Bevacizumab should be avoided for patients with squamous cell carcinoma histologic type, clinically significant hemoptysis, inadequate organ function, ECOG PS > 1, clinically significant cardiovascular disease, or medically uncontrolled hypertension. (Evidence quality: High; Strength of recommendation: Strong)
What are the most effective second-line and subsequent treatment options for patients with good performance status:	
Patients previously treated with immune checkpoint therapy without chemotherapy	3.5. Maintenance bevacizumab given with pemetrexed has no survival advantage and significantly increased toxicity compared to maintenance pemetrexed or bevacizumab alone. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	4.1. Clinicians should offer platinum-doublet chemotherapy. (Evidence quality: Low; Strength of recommendation: Strong)

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TABLE 1. Summary of All Recommendations (continued)

Category	Recommendation
Patients previously treated with chemotherapy and immune checkpoint therapy	4.2. Clinicians should offer docetaxel with or without ramucirumab if the patient has already received platinum-based chemotherapy. (Evidence quality: Low; Strength of recommendation: Strong)
	4.3. Clinicians may offer pemetrexed, nab-paclitaxel, or gemcitabine if the patient has already received platinum-based chemotherapy. (Evidence quality: Low; Strength of recommendation: Conditional)
	4.4. Clinicians may offer telisotuzumab vedotin (Teliso-V) for patients with c-MET protein–overexpressing NSCLC. (Evidence quality: Low; Strength of recommendation: Conditional)
	4.5. Clinicians may offer trastuzumab deruxtecan for patients with HER2-overexpressing NSCLC defined as HER2 IHC 3+ (by gastric scoring). (Evidence quality: Very low; Strength of recommendation: Conditional)

NOTE. The following recommendations (strong or conditional) and terminology (see guideline Data Supplement) represent reasonable options for patients depending on clinical circumstances and in the context of individual patient preferences. Recommended care should be accessible to patients whenever possible. The strength of the recommendation is defined as follows: Strong: In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects. In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects. All or almost all informed people would make the recommended choice for or against an intervention. Conditional/weak: In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists. In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists. Most informed people would choose the recommended course of action, but a substantial number would not.

Abbreviations: ECOG, Eastern Cooperative Oncology Group; HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; MET, mesenchymal epithelial transition; nab, nanoparticle albumin-bound; NGS, next-generation sequencing; NSCLC, non–small cell lung cancer; PS, performance status; TPS, tumor proportion score.

trials, most notably with ivonescimab, as detailed previously.³⁵

Recommendation 1.5.

Literature review update. In the phase III CheckMate 227 trial, patients with PD-L1 $\geq 1\%$ were randomly assigned 1:1:1 to nivolumab plus ipilimumab, nivolumab monotherapy, or chemotherapy alone, while patients with PD-L1 $< 1\%$ were randomly assigned 1:1:1 to receive nivolumab plus ipilimumab, nivolumab plus chemotherapy, or chemotherapy alone. At the 6-year landmark, with a minimum follow-up of 73.5 months, the estimated OS rate among all patients treated with nivolumab plus ipilimumab was 20% versus 11% with chemotherapy. This update did not include additional updated survival data on individual PD-L1 or histology subsets.²⁵ At the 5-year update of this study, among patients with PD-L1 $\geq 50\%$ nonsquamous NSCLC, the nivolumab plus ipilimumab arm ($n = 147$) demonstrated improved OS compared to the chemotherapy arm ($n = 137$) with a median OS of 28.4 months (95% CI, 18.0 to 44.2) and 16.7 months (95% CI, 11.4 to 23.7), respectively (HR, 0.71 [95% CI, 0.54 to 0.94]). The landmark 5-year OS for each arm was 36% and 21%, respectively.

Clinical interpretation. While the long-term data from this subset demonstrate improved OS for nivolumab plus ipilimumab compared to chemotherapy, and was also numerically higher compared to patients treated with nivolumab monotherapy, there are some important caveats. The study was not powered to compare this particular subset. To this end, the prospective phase III KEYNOTE-598 trial, which prospectively evaluated the combination of pembrolizumab plus ipilimumab versus pembrolizumab in patients with advanced NSCLC with PD-L1 $\geq 50\%$, was negative, with median OS of 21.4 months and 21.9 months, respectively.

Given this, clinicians should carefully weigh the risks and benefits of such an approach.

Recommendation 1.6.

Literature review update. The CheckMate 9LA⁴² study randomly assigned patients to nivolumab and ipilimumab plus chemotherapy versus chemotherapy. There were 174 patients in the subgroup of patients with stage IV NSCLC and high PD-L1 expression (TPS $\geq 50\%$). For these patients, nivolumab plus ipilimumab improved OS, with a HR of 0.66 (95% CI, 0.44 to 0.89). Grade 3 to 4 treatment-related AEs (TRAEs) were increased with the immunotherapy and chemotherapy intervention (in either histology or PD-L1 subgroup). At the 6-year update from this study,²³ with a median follow-up time of 75.8 months, the benefit of nivolumab plus ipilimumab plus chemotherapy was maintained in the PD-L1–high subset, with an OS HR of 0.76 (95% CI, 0.55 to 1.05), with 6-year OS rates of 17% versus 12% for chemotherapy alone.

Clinical interpretation. In patients with high symptom burden or disease burden, clinicians may choose an option of a combination of chemotherapy with immunotherapy despite a high PD-L1 value. Although there is no head-to-head comparison between the triple combination strategy (PD-1 ICI + doublet chemotherapy) with the quadruple combination strategy (PD-1 ICI + cytotoxic T-cell lymphocyte-4 [CTLA-4] ICI + doublet chemotherapy), clinicians should keep the additive toxicity of two ICIs in mind while making decisions regarding choice of treatment in these settings.

Recommendation 1.7.

Literature review update. In the POSEIDON study,⁴³ patients with untreated EGFR and ALK wild-type metastatic NSCLC were randomly assigned in a 1:1:1 fashion to receive

durvalumab plus tremelimumab plus chemotherapy, durvalumab plus chemotherapy, or chemotherapy alone. The primary end points were PFS and OS with durvalumab plus chemotherapy versus chemotherapy. PFS and OS for durvalumab plus tremelimumab and chemotherapy versus chemotherapy were secondary end points. The study met PFS and OS for the durvalumab + tremelimumab group. At the 5-year update, patients with PD-L1 $\geq 50\%$ had an improvement in OS with durvalumab plus tremelimumab plus chemotherapy versus chemotherapy alone (HR, 0.62 [95% CI, 0.45 to 0.84]). No new safety signals were identified.

Clinical interpretation. Based on the 5-year update of the POSEIDON study, durvalumab plus tremelimumab plus chemotherapy had durable OS benefit versus chemotherapy alone. This benefit was evident in all PD-L1 subgroups but remained most notable in the PD-L1–high subgroup, which is consistent with other studies involving immunotherapy in this setting. It is important to note that efficacy for durvalumab plus tremelimumab and chemotherapy was a secondary end point of this study. In addition, there are no direct comparative data against anti-PD-(L)1 monotherapy or anti-PD-(L)1 + chemotherapy to assess the differential impact of such a strategy in patients with PD-L1 $\geq 50\%$, and whether this warrants the increased toxicity risk of such an approach is uncertain. That said, durvalumab plus tremelimumab plus chemotherapy may be offered as a treatment option for driver mutation–negative, PD-L1–high NSCLC after an assessment of efficacy, AEs, and a patient–clinician discussion of the potential risks versus benefits. Intriguingly, in subset analyses evaluating survival by *KRAS*, *STK11*, and *KEAP1* status, a benefit trend was observed for durvalumab–tremelimumab combination therapy. The ongoing phase III TRITON study is prospectively evaluating the efficacy of durvalumab plus tremelimumab and chemotherapy versus pembrolizumab plus chemotherapy in this population.

PD-L1 Expression, TPS 1%–49%

Recommendations 1.8 and 1.9.

Literature review update. For patients with stage IV NSCLC, nonsquamous histology and low PD-L1 tumor expression (1%–49%), there are long-term OS data that continue to support the use of a combination of platinum-based doublet chemotherapy plus immunotherapy in the first-line advanced setting. In the 5-year update from the phase III KEYNOTE-189 study,^{37,44} among patients with PD-L1 tumor expression of 1%–49%, 5-year OS was 19.8% versus 7.7%, with a median OS of 21.8 months versus 12.1 months for pembrolizumab plus chemotherapy and placebo plus chemotherapy, respectively. In the EMPOWER-Lung 3 trial,^{36,45} at the updated 2-year follow-up and final OS analysis at 28.4 months of median follow-up, in the PD-L1 TPS 1%–49% subgroup (both histologies), the median OS was 23.2 months (95% CI, 18.1 to 27.2) for cemiplimab plus chemotherapy versus 12.0 months (95% CI, 8.3 to 15.7) for

chemotherapy alone, corresponding to a HR of 0.50 (95% CI, 0.34 to 0.74).

Clinical interpretation. The addition of immunotherapy, either pembrolizumab or cemiplimab, to carboplatin and pemetrexed is recommended for patients with metastatic NSCLC with PD-L1 of 1%–49% in the first-line setting. Long term data from landmark trials such as KEYNOTE-189 and EMPOWER-Lung 3 support this combination as a preferred option for this subgroup.

Recommendation 1.10.

Literature review update. In an updated exploratory analysis of OS in the SP142-defined PD-L1 $\geq 1\%$ subgroup from the phase III IMpower150 study,^{39,46} median OS was 24.4 months (HR, 0.71 [95% CI, 0.55 to 0.91]) in the ACP arm and 22.5 months (HR, 0.73 [95% CI, 0.57 to 0.94]) in the ABCP arm compared to the BCP arm (16 months; 95% CI, 11.2 to 20.1). Similar trends with SP263-defined TC $\geq 1\%$ biomarker-evaluable population subgroup was observed, with median OS in the ACP arm having an HR of 0.66 (95% CI, 0.5 to 0.87) and the ABCP arm having an HR of 0.69 (95% CI, 0.52 to 0.91) compared with the BCP arm. OS in the PD-L1–low subset (TC 1/2, IC 1/2) has not been reported.

Clinical interpretation. Based on these data, clinicians may offer atezolizumab plus carboplatin plus (nab)–paclitaxel $\pm$ bevacizumab as frontline therapy for advanced nonsquamous NSCLC with PD-L1 1%–49%, although the robustness of these data in this subset is somewhat weaker than reported for other studies. Studies investigating PD-1: VEGF bispecifics, most notably ivonescimab, are ongoing to determine the role for frontline VEGF-containing therapy.

Recommendation 1.11.

Literature review update. The support for the nivolumab plus ipilimumab combination in metastatic NSCLC is from the phase III randomized CheckMate 227 trial, with median OS of 17.1 versus 14.9 months (HR, 0.77 [95% CI, 0.66 to 0.91]) and 5-year OS rates at 24% versus 14% for nivolumab–ipilimumab combination versus chemotherapy, respectively, in PD-L1 $\geq 1\%$ population.^{47,48} In the PD-L1–low (1%–49%) nonsquamous subset, median OS was 16.2 months with nivolumab–ipilimumab compared to 17.5 months with chemotherapy with landmark 5-year OS of 19% and 16%, respectively.

Clinical interpretation. In the PD-L1 1%–49% nonsquamous NSCLC population, nivolumab plus ipilimumab may be considered, although the data for this subset are arguably less robust than other combinatorial strategies (with the caveat that these regimens have not been directly compared). It is important to note that, in the PD-L1–positive population enrolled to the CheckMate 227 study, patients were treated with nivolumab plus ipilimumab, nivolumab monotherapy, and chemotherapy. Therefore, there is no direct comparison between nivolumab plus ipilimumab to platinum–doublet chemotherapy plus immunotherapy in this subset. Several factors that may affect decision making in this setting include patient comorbidities, treatment

toxicity profile, and patient preference (eg, chemotherapy-sparing).

Recommendation 1.12.

Literature review update. The randomized, phase III CheckMate 9LA study evaluated first-line nivolumab plus ipilimumab with two cycles of platinum-doublet chemotherapy versus four cycles of chemotherapy in patients with metastatic NSCLC, which now has reported 6-year outcome data with continued OS benefit with nivolumab plus ipilimumab with chemotherapy over chemotherapy (HR, 0.74 [95% CI, 0.63 to 0.87]; 6-year OS rates, 16% v 10%) in the total study population. Among the subset of patients with PD-L1 1%-49% (squamous and nonsquamous histology), the median OS for nivolumab-ipilimumab-chemotherapy was 15.2 months compared to 10.4 months with chemotherapy (HR, 0.71 [95% CI, 0.54 to 0.94]).^{23,42,49}

Clinical interpretation. For patients with PD-L1 1%-49% nonsquamous NSCLC, frontline treatment with nivolumab-ipilimumab plus two cycles of platinum doublet chemotherapy may be considered. In addition to patient-related comorbidities, treatment side effects, and patient-physician preference, an additional consideration for this regimen includes the status of molecular mutations associated with single-agent immunotherapy resistance, namely *STK11* and *KEAP1*. Small subset data from the CheckMate 9LA study report a signal of benefit for this regimen in patients with this molecular profile, and this is further supported by translational investigation.⁵⁰ Prospective phase trials (such as the phase III TRITON study) are needed to confirm the efficacy of a CTLA-4-containing regimen in this population.

Recommendation 1.13.

Literature review update. In the phase III POSEIDON trial,⁵¹ publication of the 5-year OS outcomes continued to demonstrate an OS benefit for durvalumab plus tremelimumab and chemotherapy compared to chemotherapy (HR, 0.76 [95% CI, 0.64 to 0.89]).⁴³ Among patients with nonsquamous NSCLC and PD-L1 $\geq 1\%$, the OS HR was 0.63 (95% CI, 0.48 to 0.83) favoring the durvalumab-tremelimumab arm. The PD-L1 TC 1%-49% subgroup (squamous + nonsquamous histologies), which was not a prespecified OS analysis, showed a HR of 0.81 (95% CI, 0.61 to 1.08) favoring the durvalumab-tremelimumab-chemotherapy combination.

Clinical interpretation. The combination of durvalumab plus tremelimumab and platinum-doublet chemotherapy may be considered for nonsquamous NSCLC with PD-L1 1%-49%. As previously stated, clinicians should be aware that the durvalumab-tremelimumab combination arm was a key secondary end point, not primary end point of the POSEIDON trial. A unique aspect of this study was the potential signal of benefit seen for durvalumab-tremelimumab combination therapy in patients with mutations in *KRAS*, *STK11*, or *KEAP1*. That said, this represented a small subset of the overall study population that was not prespecified. The ongoing phase III

TRITON study will help to determine whether a CTLA-4-containing regimen is superior in advanced NSCLC with this molecular signature.

Recommendation 1.14.

Literature review update. The phase III KEYNOTE-042 study evaluated pembrolizumab versus platinum-doublet chemotherapy in patients with advanced NSCLC with PD-L1 TPS $\geq 1\%$. The study met its primary OS end point, and at the 5-year update, continued to demonstrate an OS benefit for pembrolizumab compared to chemotherapy (median OS 16.4 v 12.1 months; HR, 0.79 [95% CI, 0.70 to 0.89]). In an exploratory analysis of PD-L1 TPS 1%-49%, OS for pembrolizumab versus chemotherapy HR was 0.88 (95% CI, 0.75 to 1.04) corresponding to an estimated 5-year OS rate at 11.9% for pembrolizumab group versus 7.4% for the chemotherapy group.^{52,53}

Clinical interpretation. Based on results of the phase III KEYNOTE-042 trial, pembrolizumab received FDA approval for advanced NSCLC with PD-L1 TPS $\geq 1\%$. That said, the efficacy benefit appeared to be largely driven by the PD-L1-high ($\geq 50\%$) subset. Because of this, the panel would recommend that pembrolizumab monotherapy be primarily considered for patients with PD-L1 1%-49% who have a contraindication to or decline combination therapy.

Unknown or Negative PD-L1 Expression, TPS 0%

Recommendations 1.15 and 1.16.

Literature review update. At the 5-year update from KEYNOTE-189, patients with advanced nonsquamous NSCLC with PD-L1 $< 1\%$ treated with pembrolizumab plus chemotherapy (n = 127) had a median OS of 17.2 months (95% CI, 13.8 to 22.8) compared to 10.2 months (95% CI, 7.0 to 13.5) in those treated with chemotherapy (n = 63), HR, 0.55 (95% CI, 0.39 to 0.76). The landmark 5-year OS for each group was 9.6% and 5.3%, respectively.³⁷

In EMPOWER-Lung 3, extended follow-up beyond 2 years and for specific PD-L1 tertiles separated by disease histology have yet to be reported. At the 2-year follow-up, among patients with advanced NSCLC with PD-L1 $< 1\%$ (both histologies), the median OS was 12.8 months (95% CI, 9.6 to 15.6) compared to 14.2 months with chemotherapy (95% CI, 9.1 to 18.0), HR, 0.73 (95% CI, 0.50 to 1.08).³⁶

Clinical interpretation. Patients with nonsquamous NSCLC may derive benefit from single or double ICI regimens in combination with chemotherapy, although OS outcomes are similar. Regimens with chemotherapy plus ICI singlet may offer less toxicity with equivalent survival benefit. Subgroup analyses for cemiplimab may have been underpowered to detect a statistically significant OS benefit. Furthermore, the lack of reporting of these subsets by disease histology, while likely limited by sample size, makes interpretation of these data for individual disease histologies challenging.

Recommendation 1.17.

Literature review update. The final OS analysis of IMpower150 demonstrated no significant difference between the ACP group (n = 164), the ABCP group (n = 167), and the BCP group (n = 173; ACP v BCP HR, 0.96 [95% CI, 0.76 to 1.22]; ABCP v BCP HR, 0.90 [95% CI, 0.71 to 1.14]) for patients with advanced PD-L1-negative (TCO and ICo) nonsquamous NSCLC. The median OS was 14.8 versus 16.9 versus 14.1 months in the ACP versus ABCP versus BCP groups, respectively. The AEs in the ABCP group (41.2%) were substantially higher than that of the ACP group (14.5%) and the BCP group (26.4%).^{39,46}

In the phase III IMpower130 trial, among patients with PD-L1-negative nonsquamous NSCLC without driver alterations, there was a trend toward benefit in the atezolizumab plus carboplatin and nab-paclitaxel group (n = 235) compared to the carboplatin plus nab-paclitaxel group (n = 121), with median OS of 15.2 months and 12.0 months, respectively (HR, 0.81 [95% CI, 0.61 to 1.08]).

Clinical interpretation. For patients with PD-L1-negative nonsquamous NSCLC treated in the IMpower150 study, the addition of bevacizumab to a platinum doublet chemotherapy regimen with atezolizumab did not significantly improve OS. The addition of bevacizumab increases AEs and should be used with caution in subgroup populations such as older adults. An OS benefit trend was noted for the addition of atezolizumab to carboplatin plus nab-paclitaxel in the IMpower130 study.

Recommendation 1.18.

Literature review update. The Checkmate 9LA regimen demonstrated a landmark 6-year OS rate of 20% among patients with PD-L1-negative NSCLC compared to 7% with chemotherapy alone. For patients with PD-L1-negative nonsquamous NSCLC, the median OS was 18.6 months (95% CI, 13.2 to 22.7) in the nivolumab + ipilimumab + chemotherapy group versus 12.4 months (95% CI, 7.7 to 15.2) in the chemotherapy only group, HR, 0.69 (95% CI, 0.51 to 0.94).^{23,42}

Clinical interpretation. The addition of a limited use of chemotherapy in addition to a doublet ICI regimen demonstrated significant and durable long-term benefits among patients with PD-L1-negative NSCLC, suggesting this is a reasonable option to consider in this patient subset. That said, due to the increase in treatment-related toxicities seen with combination checkpoint blockade, updated risk stratification and guidelines for selection of patients to receive this treatment regimen upfront are needed.²³

Recommendation 1.19.

Literature review update. For patients with PD-L1-negative nonsquamous NSCLC who received nivolumab + ipilimumab (n = 141) versus nivolumab + chemotherapy (n = 134) versus chemotherapy (n = 140), OS was improved in the nivolumab + ipilimumab subgroup when compared to chemotherapy: HR, 0.70; 95% CI, 0.54 to 0.90, at the 5-year update. OS was also improved in the nivolumab +

chemotherapy group versus chemotherapy alone (HR, 0.77 [95% CI, 0.60 to 0.99]). The nivolumab + ipilimumab group was not compared to the nivolumab + chemotherapy group. The median OS was 17.5 months (95% CI, 12.3 to 23.9) in the nivolumab + ipilimumab group versus 17.7 months (95% CI, 13.5 to 21.6) in the nivolumab + chemotherapy group versus 13.1 months (95% CI, 9.8 to 15.3) in the chemotherapy alone group.^{47,48} The landmark 5-year OS with nivolumab + ipilimumab was 19% compared to 12% with nivolumab plus chemotherapy and 8% with chemotherapy.⁵⁴

Clinical interpretation. Nivolumab plus ipilimumab demonstrated a durable survival benefit for patients with PD-L1-negative nonsquamous NSCLC compared to chemotherapy alone. This regimen may be considered in this subset, particularly for patients who may be ineligible to receive platinum-doublet chemotherapy.

Recommendation 1.20.

Literature review update. Among 368 patients with advanced NSCLC treated in the phase III POSEIDON study with PD-L1-negative tumors, the addition of tremelimumab to durvalumab and chemotherapy improved OS (HR, 0.77 [95% CI, 0.58 to 1.00]). The 5-year OS was 15.7% in the tremelimumab-durvalumab-chemotherapy combination group versus 13.0% in the durvalumab-chemotherapy group versus 6.8% in the chemotherapy group. For patients with nonsquamous NSCLC and PD-L1 <1%, OS benefit was not statistically significant compared to chemotherapy (HR, 0.81 [95% CI, 0.59 to 1.13]).^{43,51}

Clinical interpretation. The combination of tremelimumab-durvalumab-chemotherapy versus chemotherapy did not result in a statistically significant improvement in landmark 5-years OS for patients with nonsquamous NSCLC and PD-L1 <1% tumors. That said, conclusions drawn on subset data can be challenging, and whether or not other molecular features such as *KEAP1* or *STK11* mutations might predict benefit with this regimen is an ongoing question.

Squamous Cell Carcinoma**PD-L1 Expression, TPS ≥50%****Recommendation 2.1.**

Literature review update. Three open-label, phase III randomized trials have evaluated PD-(L)1 monotherapy versus chemotherapy in patient with tumor expression of PD-L1 ≥50%. These include the IMpower110 (N = 205; atezolizumab),²⁸ KEYNOTE-024 (N = 305; pembrolizumab),³⁰ and EMPOWER-Lung 1 (N = 563; cemiplimab) trials.³¹ The results from these studies showed consistent significant improvement in both PFS and OS for patients who received ICI compared to chemotherapy with IMpower110 reporting median OS of 20.2 months (95% CI, 17.2 to 27.9) for atezolizumab in the PD-L1-high subset; KEYNOTE-024 reporting 26.3 months (95% CI, 18.3 to 40.4) for pembrolizumab; and EMPOWER-Lung 1 reporting 26.1 months (95% CI, 22.1 to 31.9) for cemiplimab. Both the EMPOWER-

Lung 1 and KEYNOTE-024 studies have reported 5-year OS updates, demonstrating a 5-year OS of 29.0% for cemiplimab and 31.9% for pembrolizumab. In EMPOWER-Lung 1, among 245 patients with squamous NSCLC, at 5 years, the median OS was 22.7 months (95% CI, 17.3 to 31.5) versus 13.5 months (95% CI, 10.0 to 16.2) with chemotherapy (HR, 0.509 [95% CI 0.376 to 0.689]).

Clinical interpretation. The proportion of patients with squamous NSCLC treated with ICI in each study varied: 18.8% in KEYNOTE-024, 25.2% in IMpower110, and 43.3% in EMPOWER-Lung 1. The OS benefit of single-agent PD-(L)1 ICI in this patient population compared to chemotherapy is clear. While there are retrospective analyses that suggest both clinical presentation and molecular markers might be considered for intensifying therapy, such as quantity and location of disease burden at diagnosis, and tumor mutations such as *STK11*, *KEAP1*, *SMARCA4*, and *CDN2KA*, prospective analyses for definitive biomarkers for consideration in this space are ongoing. Due to the lack of prospective studies in this space and recognizing the increased rates of AEs with addition of chemotherapy, PD-(L)1 monotherapy is the generally preferred approach in the setting of tumor PD-L1 TPS $\geq 50\%$. KEYNOTE-598, the phase III study evaluating addition of anti-CTLA-4 ICI to pembrolizumab in this patient population, was stopped early due to lack of efficacy and increased toxicity, showing this is not an appropriate treatment for all patients with tumor PD-L1 TPS $\geq 50\%$. The phase III HARMONI-2 study in China evaluating the VEGF-PD-1 bispecific antibody, ivonescimab, showed improved PFS and encouraging improvement in OS compared to pembrolizumab, which has led to its approval in China, and global studies in NSCLC are underway to evaluate if this dual targeting agent significantly improves survival.

Recommendation 2.2.

Literature review update. Updated analyses of the phase III KEYNOTE-407 evaluating pembrolizumab plus platinum doublet chemotherapy compared to chemotherapy alone showed a continued significant improvement in OS, with a median OS of 19.9 months (95% CI, 12.2 to 25.2) with pembrolizumab (n = 73) compared to 11.5 months (95% CI, 7.5 to 17.1) with chemotherapy (n = 73; HR, 0.68 [95% CI, 0.47 to 0.97]) among patients with PD-L1 TPS $\geq 50\%$ squamous NSCLC.⁵⁵ The 5-year OS rate was 23.3% with pembrolizumab plus chemotherapy and 8.3% with chemotherapy.⁵⁵ Updated analysis of the phase III EMPOWER-Lung 3 study evaluating platinum doublet plus cemiplimab revealed continued significant improvement in OS (median OS, 21.1 months) across all tumor histologies and PD-L1 expression levels; exploratory subgroup analyses revealed median OS of 22.2 months (HR, 0.77 [95% CI, 0.40 to 1.45]) for patients with PD-L1 $\geq 50\%$ squamous NSCLC.³⁶

Clinical interpretation. Both KEYNOTE-407 and EMPOWER-Lung 3 were phase III placebo-controlled studies evaluating combination chemotherapy + PD-1 ICI in all PD-L1 subsets. While the subset analyses in each trial showed

benefit for squamous tumors with high PD-L1 expression, both trials showed increased rates of all grade and high-grade AEs. With the current lack of prospective head-to-head comparison, expert consensus is to consider addition of cytotoxic chemotherapy if there is a significant tumor burden or risk for visceral crisis with delayed response to therapy, as there are increased response rates with these combinatorial approaches (KEYNOTE 407 PD-L1-high subset ORR, 64.4% compared to KEYNOTE-024 monotherapy ORR, 46.1%).

Recommendation 2.3.

Literature review update. The updated analyses of the phase III CheckMate227 trial evaluating nivolumab + ipilimumab compared to nivolumab monotherapy or chemotherapy alone showed sustained benefit of a dual ICI approach. Among patients with PD-L1 $\geq 50\%$ squamous NSCLC, the nivolumab plus ipilimumab arm (n = 58) demonstrated improved OS compared to the chemotherapy arm (n = 55) with a median OS of 14.9 months (95% CI, 10.3 to 22.2) compared to 8.3 months (95% CI, 5.6 to 14.5). The landmark 5-year OS for each arm was 23% and 8%, respectively.⁴⁷

Clinical interpretation. While there is significant benefit for using ICI in PD-L1-high squamous NSCLC, it is unclear what the absolute benefit is for adding CTLA-4 ICI to PD-1 ICI for these patients. Specifically, the KEYNOTE-598 study, which evaluated the addition of ipilimumab to pembrolizumab, had to be stopped early due to lack of efficacy and increased toxicity. In CheckMate 227, the nivolumab monotherapy arm had similar OS data compared to dual ICI (median OS 15.2 months [95% CI, 0.2 to 26.4]; landmark 5-year OS 23%). While there are several retrospective analyses and ongoing studies in molecular subsets of NSCLC suggesting potential benefit of CTLA-4, at this time, prospective data are lacking to suggest that addition of CTLA-4 ICI to PD-1 ICI offers improved outcomes for our patients with PD-L1 TPS $\geq 50\%$ squamous NSCLC.

Recommendation 2.4.

Literature review update. In the updated analysis of CheckMate-9LA where patients were randomized to nivolumab + ipilimumab + chemotherapy compared to chemotherapy alone, significant improvement was seen in both high PD-L1 and squamous histology subset analyses. For tumors with PD-L1 TPS $\geq 50\%$, median OS with combination therapy was 18.9 months (95% CI, 13.1 to 29.1) compared to 12.9 months (95% CI, 9.4 to 17.6) with chemotherapy (HR, 0.76 [95% CI, 0.55 to 1.05]). The 6-year OS was 17% with combination therapy compared to 12% with chemotherapy alone; for PD-L1 $\geq 1\%$ squamous tumors, median OS with combination therapy was 14.2 months (95% CI, 12.4 to 23.3) compared to 10.5 months (95% CI, 7.1 to 14.1) with chemotherapy (HR, 0.70 [95% CI, 0.50 to 0.99]). The 6-year OS with combination therapy was 13% compared to 5% with chemotherapy alone. 92% of patients on the quadruplet therapy arm (n = 358) compared to 88% in the chemotherapy arm (n = 349) experienced TRAEs, with rates of 48% and 38% for high-grade TRAEs, respectively.²³

Clinical interpretation. Dual ICI plus chemotherapy is an option for patients with metastatic NSCLC. However, as the data for efficacy with single-agent PD-(L)1 are strong, and the toxicity data of quadruplet therapy show significant impact for patients, we await prospective analyses to confirm which patients with PD-L1 TPS $\geq 50\%$ squamous NSCLC will benefit from the addition of CTLA-4. Specifically, rates of TRAEs and high-grade TRAEs are higher with this quadruplet therapy than in the chemotherapy arm and in studies of single-agent PD-(L)1 ICI, suggesting toxicity profiles must be carefully considered as there is no clear comparison of regimens to suggest potential benefit for all patients with PD-L1-high squamous NSCLC. In the setting of high disease burden, combination chemotherapy and ICI should be considered; the absolute benefit of adding CTLA-4 ICI to PD-1 ICI in this clinical scenario is unknown based on current data. Again, ongoing biomarker studies to evaluate optimized use of anti-CTLA-4 in metastatic PD-L1-high NSCLC are ongoing.

Recommendation 2.5.

Literature review update. The phase III POSEIDON trial evaluating durvalumab + tremelimumab + platinum-doublet chemotherapy compared to durvalumab + chemotherapy compared to chemotherapy alone continued to show improved OS for quadruplet therapy with sustained OS benefit versus chemotherapy (HR, 0.76 [95% CI, 0.64 to 0.89]) and 5-year OS: 15.7% versus 6.8%. At the 5-year update, subset analyses showed patients with tumor PD-L1 TPS $\geq 50\%$ had an improvement in OS with durvalumab plus tremelimumab plus chemotherapy versus chemotherapy alone (HR, 0.62 [95% CI, 0.45 to 0.84]). For patients with squamous histology tumors, OS HR was 0.85 (95% CI, 0.65 to 1.10), favoring durvalumab plus tremelimumab and chemotherapy.⁴³ Among patients with squamous histology and PD-L1 TPS $\geq 1\%$, the OS HR was 0.89 (95% CI, 0.64 to 1.25).

Clinical interpretation. Similar to CheckMate 9LA, the benefits of quadruplet therapy in POSEIDON suggest this is an option for patients. While the benefit seems most apparent in high PD-L1 disease, it is unclear if this benefit is worth the potential increased toxicity compared to using PD-(L)1 ICI, as there are no prospective data to suggest an increased absolute survival benefit, particularly in patients with squamous histology, in whom the benefit appears to be more modest.

PD-L1 Expression TPS, 1%–49%

Recommendations 2.6 and 2.7.

Literature review update. Outcomes from two large, phase III, RCTs support the use of combination of platinum and taxane-based chemotherapy regimens with the ICIs, pembrolizumab or cemiplimab, for the first-line treatment of advanced, squamous cell carcinoma of the lung with PD-L1 expression of 1%–49% TPS.^{45,56} In the 5-year update from

the KEYNOTE-407 trial, among 207 patients with PD-L1 TPS 1%–49%, the 5-year OS rates were 20.6% for pembrolizumab combined with chemotherapy versus 7.6% for chemotherapy alone, with a HR of 0.61 (95% CI, 0.45 to 0.83).⁵⁵ The median OS for the PD-L1 TPS 1%–49% group was 18.0 months (95% CI, 13.6 to 22.8) for pembrolizumab combined with chemotherapy versus 13.1 months (95% CI, 9.1 to 15.2) for chemotherapy alone. Similarly, the 2-year update of the phase III EMPOWER-Lung 3 Part 2 trial continues to demonstrate significant improvement in outcomes with the addition of cemiplimab to chemotherapy among patients with advanced NSCLC.³⁶ In a subgroup analysis of 81 patients with squamous histology and PD-L1 TPS 1%–49%, the median OS was 23.2 months with the combination of cemiplimab plus chemotherapy versus 8.6 months for chemotherapy alone (HR, 0.52 [95% CI, 0.29 to 0.92]).³⁶

Clinical interpretation. The addition of either pembrolizumab or cemiplimab to carboplatin and taxane-based chemotherapy for four cycles followed by maintenance ICI is recommended for the first-line treatment of advanced squamous NSCLC with PD-L1 TPS of 1%–49%. Long-term follow-up data from the landmark trial KEYNOTE-407 and a subgroup analysis of patients with squamous NSCLC from the phase III EMPOWER-Lung 3 trial support the efficacy and safety of this combination as a preferred option for this subgroup.

Recommendation 2.8.

Literature review update. The evidence supporting the use of the combination of nivolumab and ipilimumab in advanced NSCLC comes from the phase III CheckMate 227 trial.⁴⁸ At the 5-year update, among patients with PD-L1 $\geq 1\%$ squamous NSCLC, the combination of nivolumab plus ipilimumab showed a durable OS benefit (HR, 0.69 [95% CI, 0.52 to 0.91]). However, the magnitude of benefit observed among patients with PD-L1 TPS 1%–49% was less pronounced. Among 181 patients with squamous histology and PD-L1 TPS 1%–49% (n = 60 received nivolumab plus ipilimumab, n = 58 received nivolumab alone, and n = 63 received chemotherapy alone), the median OS with nivolumab plus ipilimumab was 14.8 months (95% CI, 8.6 to 18.7) versus 10.6 months (95% CI, 7.8 to 14.1) with chemotherapy alone (HR, 0.76 [95% CI, 0.52 to 1.10]).⁴⁷ The landmark 5-year OS in each group was 8% and 2%, respectively.

Clinical interpretation. Updated outcome data from the CheckMate 227 trial confirm dual CTLA-4 and PD-L1 inhibition as an option for the frontline management of NSCLC. However, the addition of the CTLA-4 inhibition increases toxicity rates, and in the absence of direct head-to-head data, it is difficult to gauge the added benefit of CTLA-4 inhibition when compared to chemotherapy plus PD-L1, where subset data show durable clinical efficacy in the management of squamous NSCLC with PD-L1 1%–49%. Additional work is needed to determine if there are certain patient subgroups with PD-L1 1%–49% squamous NSCLC that may benefit from a CTLA-4-containing regimen.

Recommendation 2.9.

Literature review update. The evidence supporting the use of the combination of nivolumab and ipilimumab with two cycles of platinum-based chemotherapy comes from the phase III CheckMate 9LA trial.^{42,49} The 6-year and final outcomes from CheckMate 9LA confirmed a durable OS benefit for nivolumab plus ipilimumab with chemotherapy.²³ Among a subgroup analysis of patients with PD-L1 1%-49% (squamous and nonsquamous), the observed median OS was 15.2 months (95% CI, 12.6 to 21.2) for the combination versus 10.4 months (95% CI, 8.7 to 12.4) for chemotherapy alone (HR, 0.71 [95% CI, 0.54 to 0.94]). This OS benefit was sustained in patients with squamous histology and PD-L1 expression $\geq 1\%$ (median OS 14.2 months [95% CI, 12.4 to 23.2] for combination versus 10.5 months [95% CI, 7.1 to 14.1] for chemotherapy; HR, 0.70 [95% CI, 0.50 to 0.99]). Grade 3 and 4 TRAEs were higher in the nivolumab plus ipilimumab with chemotherapy group (48% for combination v 38% for chemotherapy) and TRAEs leading to discontinuation of any treatment component were also higher in the combination group (22% for combination v 9% for chemotherapy).⁴⁹ Notably, among patients who discontinued all components of the combination arm ($n = 61$, 17%), 22% remained alive and treatment-free at 5 years, and the 5-year OS rate was 37%.

Clinical interpretation. Updated results from CheckMate 9LA confirm the feasibility and efficacy of combination chemotherapy with dual CTLA-4 and PD-L1 inhibition. Despite lowering the number of chemotherapy cycles, CheckMate 9LA showed increased grade 3 to 4 TRAEs associated with dual checkpoint inhibition and a significant increase in treatment discontinuation rates. The benefit of dual checkpoint inhibition compared to other approved regimens in squamous NSCLC with PD-L1 1%-49% remains unclear, while the increased toxicity cost is consistent with other dual checkpoint inhibition trials. Notably, despite the increased toxicity, even among patients who discontinued therapy, prolonged responses were observed with 22% of participants remaining alive at 5 years.

Recommendation 2.10.

Literature review update. In the phase III POSEIDON trial, at the 5-year updated analysis, among patients with PD-L1 1%-49% (squamous and nonsquamous histology), OS favored the durvalumab plus tremelimumab and chemotherapy arm compared to chemotherapy alone with an OS HR of 0.81 (95% CI, 0.61 to 1.08). Among patients with squamous histology and PD-L1 $\geq 1\%$, the OS HR for combination therapy compared to chemotherapy alone was 0.89 (95% CI, 0.64 to 1.25). In the total study population, grade 3 and 4 TRAEs were prominent, affecting 51.8% (durvalumab + tremelimumab + chemotherapy), 44.6% (durvalumab + chemotherapy), and 44% (chemotherapy alone).

Clinical interpretation. While the combination of durvalumab plus tremelimumab and chemotherapy is feasible in the first-line setting for patients with advanced NSCLC

without actionable mutations, the increased toxicity of combination immune checkpoint blockade, coupled with quite modest long-term efficacy data in patients with squamous NSCLC, suggests further research is needed to identify patient subsets most likely to benefit from this approach.

Recommendation 2.11.

Literature review update. The 5-year update of the phase III KEYNOTE-042 trial demonstrated an OS benefit favoring pembrolizumab for advanced PD-L1-positive NSCLC regardless of PD-L1 expression level, with a 5-year OS of 16.6% (95% CI, 13.7% to 19.6%) compared to 8.5% (95% CI, 6.4% to 11%) with chemotherapy.⁵³ A subset analysis confirmed this OS benefit is maintained in patients with squamous histology (HR, 0.76 [95% CI, 0.63 to 0.92]). However, in the specific subset of patients with a PD-L1 TPS of 1%-49% (squamous + nonsquamous histology), the increase in median OS with pembrolizumab (13.4 months [95% CI, 10.7 to 16.9 months]) versus chemotherapy (12.1 months [95% CI, 11.0 to 14.0 months]) was more modest (HR, 0.88 [95% CI, 0.75 to 1.04]).⁵²

Clinical interpretation. Treatment with single-agent pembrolizumab in patients with marginal performance status (PS), who are ineligible for chemotherapy, or who decline chemotherapy, is an appropriate consideration with close monitoring for response for patients with squamous NSCLC with PD-L1 1%-49%. That said, given that the survival benefit appeared to be largely driven by the PD-L1 $\geq 50\%$ subset in the KEYNOTE-042 study, pembrolizumab monotherapy should primarily be considered only for patients who are not eligible for or decline combination therapy strategies.

Unknown or Negative PD-L1 Expression, TPS 0%

Recommendations 2.12 and 2.13.

Literature review update. In the phase III KEYNOTE 407 trial,^{55,56} approximately a third (34%) of patients had PD-L1-negative tumors. At the initial publication, an OS and PFS benefit was observed in this subset, with a median OS of 15.9 versus 10.2 months (HR, 0.61 [95% CI, 0.38 to 0.98]) and median PFS of 6.3 versus 5.3 months (HR, 0.68 [95% CI, 0.47 to 0.98]). At the 5-year update, however, there was no statistically significant improvement in OS in patients with PD-L1-negative tumors treated with pembrolizumab versus placebo plus chemotherapy, with a 5-year OS rate of 10% for pembrolizumab and 13% for placebo (HR, 0.83 [95% CI, 0.61 to 1.13]).⁵⁵

In the phase III EMPOWER-Lung 3 trial,^{36,45} while benefit was seen for cemiplimab plus chemotherapy compared to chemotherapy alone in the squamous cell subgroup (HR, 0.56 [95% CI, 0.37 to 0.84]), there was a lack of benefit observed in the TPS $< 1\%$ subgroup (HR, 1.01 [95% CI, 0.63 to 1.60]). That said, this subset included both squamous and nonsquamous histologies. In the 2-year follow-up analysis,

when looking specifically at the PD-L1–negative squamous NSCLC subset ($n = 54$), an OS trend was observed with cemiplimab plus chemotherapy compared to chemotherapy (median OS 21.9 v 16.7 months; HR, 0.60 [95% CI, 0.30 to 1.20]).

Clinical interpretation. Evaluation of the subgroup analysis for patients with unknown or TPS <1% must be interpreted with caution. While an initial OS signal of benefit for carboplatin–taxane–pembrolizumab was observed in the KEYNOTE 407 study, this appeared to wane with time and was not evident at the 5-year update. On the contrary, in a limited, 54-patient subset from EMPOWER–Lung 3, a trend of benefit was noted at the 2-year update for the squamous PD-L1–negative population. While additional prospective studies in this subset are surely needed, the panel continues to recommend that a combination of pembrolizumab with carboplatin and paclitaxel (or nab–paclitaxel) or cemiplimab with platinum and paclitaxel may be offered as a treatment option in this subset.

Recommendation 2.14.

Literature review update. In the phase III Checkmate 9LA clinical trial, an OS benefit was reported for nivolumab plus ipilimumab and chemotherapy versus chemotherapy in both the squamous cell population (median OS 14.5 v 9.1 months; HR, 0.62 [95% CI, 0.45 to 0.86]) and the PD-L1 <1% population (median OS 16.8 v 9.8 months; HR, 0.62 [95% CI, 0.45 to 0.85]). At the 6-year update, among 72 patients with PD-L1–negative squamous NSCLC, a signal of benefit was observed with nivolumab–ipilimumab–chemotherapy with a median OS of 15.3 months compared to 8.0 months with chemotherapy (HR, 0.52 [95% CI, 0.31 to 0.86]) and landmark 6-year OS of 19% and 6%, respectively.^{23,42}

Clinical interpretation. While encouraging, these data should be interpreted with caution. There was no stratification of squamous and adenocarcinoma histology, so it is unclear how much of this benefit can be seen in squamous cell patients. Furthermore, the JCOG2007 trial,⁵⁷ which compared the CheckMate 9LA regimen to pembrolizumab plus chemotherapy, was stopped early due to high rates of treatment–related deaths in the nivolumab plus ipilimumab and chemotherapy arm. Subgroup analysis did not reveal any specific population that benefited from combination nivolumab–ipilimumab compared to pembrolizumab, including in patients with squamous histology (HR, 1.02 [95% CI, 0.47 to 2.21]) or patients with PD-L1–negative tumors (HR, 1.44 [95% CI, 0.82 to 2.50]); however, a dedicated analysis of patients with squamous histology and PD-L1–negative tumors was not performed.

However, given the durable clinical benefit seen across all subgroups at 6 years in the CheckMate 9LA trial, including squamous histology regardless of PD-L1 status and all PD-L1 levels regardless of histology, it is reasonable to consider this regimen as an option within the squamous cell PD-L1 <1% population.

Recommendation 2.15.

Literature review update. In the phase III CheckMate 227 trial, while the primary end point explored the efficacy of nivolumab plus ipilimumab in patients with advanced PD-L1–positive NSCLC, an OS benefit was also noted for those with PD-L1 <1%. In the original data publication, among 92 patients with squamous NSCLC with PD-L1 <1%, the median OS with nivolumab–ipilimumab was 15.9 months compared to 8.5 months with chemotherapy (HR, 0.49 [95% CI, 0.30 to 0.79]). At the 5-year update, this benefit was maintained, with a HR of 0.52 (95% CI, 0.34 to 0.82) favoring nivolumab–ipilimumab, and landmark 5-year OS of 18% compared to 4% with chemotherapy.^{47,58}

Clinical interpretation. While not a primary end point of the study, a strong efficacy signal of benefit was observed for patients with PD-L1–negative NSCLC in the phase III CheckMate 227 trial. In the squamous histology subset, this signal was maintained, although this should be interpreted with caution, given small patient numbers. That said, nivolumab plus ipilimumab is a reasonable option that may be considered in this patient subset.

Recommendation 2.16.

Literature review update. In the phase III POSEIDON study, at initial data release, a signal of benefit was seen in advanced NSCLC with PD-L1 <1% for durvalumab plus tremelimumab and chemotherapy compared to chemotherapy (HR, 0.77 [95% CI, 0.58 to 1.00]). This signal was less clear in patients with squamous histology, with a HR of 0.88 (95% CI, 0.68 to 1.16) favoring the durvalumab–tremelimumab arm. At the 5-year update of this study, among patients with squamous NSCLC histology and PD-L1 <1%, this modest signal continued to be observed with an OS HR of 0.80 (95% CI, 0.53 to 1.23) favoring the durvalumab–tremelimumab combination arm.⁵¹

Clinical interpretation. The results of the POSEIDON trial are challenging to interpret, given the reporting and subsequent approval of a cohort analyzed as a key secondary end point. Drawing conclusions from a subset of this cohort can be even more challenging. These data raise questions about the extent of benefit in the squamous cell NSCLC population that is PD-L1–negative. Additional work is needed to explore whether certain patient subsets, most notably those with *STK11* or *KEAP1* mutations, may benefit from this approach. With the limitations of these data in mind, this regimen may be considered for patients with advanced PD-L1–negative squamous NSCLC.

General Approaches

Recommendation 3.1.

Literature review update. Early integration of interdisciplinary palliative care into the management of advanced NSCLC has demonstrated consistent benefits across multiple randomized trials and meta-analyses. The pivotal RCT by

Temel et al⁵⁹ enrolled ambulatory patients with newly diagnosed metastatic NSCLC. Patients receiving early outpatient palliative care, initiated within 8 weeks of diagnosis, experienced significantly improved quality of life (Functional Assessment of Cancer Therapy-Lung), fewer depressive symptoms (16% v 38%), and longer median survival (11.6 v 8.9 months; $P = .02$), despite receiving less aggressive end-of-life care.

A 2024 randomized multisite trial directly compared early palliative care delivered via telehealth versus in-person outpatient visits. At 24 weeks, quality-of-life outcomes were equivalent between groups (difference, 2.0 [90% CI, 0.1 to 3.9]), indicating telehealth is a viable and effective alternative for expanding access.⁶⁰

A large randomized “stepped” palliative care trial ($N = 507$) demonstrated that a needs-triggered model (fewer baseline visits, escalation when quality of life declines) was noninferior in quality-of-life outcomes compared to standard monthly visits, while conserving resources highlighting scalable delivery models without loss of benefit.⁶¹

Systematic reviews and meta-analyses have repeatedly shown that early palliative care improves patient-reported outcomes, decreases health care utilization, and does not shorten survival, often prolonging it.^{62,63}

Clinical interpretation. Evidence is robust and consistent that early referral to interdisciplinary palliative care should occur shortly after diagnosis of advanced lung cancer, regardless of histologic subtype or presence of driver mutations. The recommendation strongly endorses concurrent palliative care alongside disease-directed treatment.

Virtual delivery (telehealth) of palliative care is an acceptable and effective alternative to in-person care and should be utilized where in-person access is limited. A stepped, patient-reported outcome triggered model can be employed in resource-constrained settings to ensure efficiency without compromising outcomes.

When feasible, institutional protocols should encourage palliative care referral near diagnosis, with standardized documentation of symptom assessment, goals-of-care discussions, and regular follow-up. Such integration can reduce aggressive end-of-life care, improve quality of life, and may even extend survival.

Recommendation 3.2.

Literature review update. Platinum-based doublet chemotherapy remains the backbone of systemic therapy for advanced NSCLC in patients who are ineligible for ICIs due to contraindications such as active autoimmune disease, solid organ transplantation, or history of severe immune-related AEs.

The efficacy of platinum doublet chemotherapy has been well established through numerous randomized trials pre-dating the ICI era. Historical studies, including ECOG 1594 and others, demonstrated similar OS among cisplatin- and carboplatin-based doublets when combined with third-generation agents such as paclitaxel, gemcitabine, or docetaxel. Median OS typically ranged between 8 and 10 months, with response rates around 20%–30%.

Histology-driven selection of doublet agents has improved tolerability and outcomes. For nonsquamous NSCLC, the combination of cisplatin or carboplatin with pemetrexed is associated with superior survival and reduced toxicity compared to gemcitabine or paclitaxel combinations, as shown in the phase III JMDB trial.⁶⁴ For squamous histology, combinations with gemcitabine or taxanes remain standard.

Many retrospective real-world studies showed that the use of immunotherapy in patients with cancer and history of autoimmune disorders led to higher rate of immune-related AEs. However, their occurrence was not related to improved outcomes. These studies showed that patients with preexisting autoimmune diseases that are non-life-threatening, non-neurologic, and under good control can be considered for initiation of immunotherapy under the care of a multidisciplinary team.^{65,66}

Clinical interpretation. For patients with advanced NSCLC who are not eligible for ICIs, platinum-based doublet chemotherapy remains the standard of care, provided that PS is preserved (ECOG PS 0–1, or selected 2). This recommendation is grounded in high-quality evidence from pre-ICI era randomized trials and supported by consistent real-world outcomes in ICI-ineligible cohorts, as well as the control arm from various more recent pivotal phase III trials.

Efforts to improve the safety and accessibility of systemic therapy in these populations should continue, including evaluation of alternative immunotherapy agents with reduced autoimmune risk or targeted agents where applicable. However, platinum doublets remain a critical therapeutic option and should be offered when appropriate.

Recommendation 3.3.

Literature review update. Non-platinum-based chemotherapy regimens have demonstrated reasonable efficacy in patients with advanced NSCLC. A phase III trial comparing the non-platinum-based doublet of gemcitabine plus docetaxel with cisplatin plus vinorelbine demonstrated similar PFS, OS, and ORR with a lower rate of toxicity with the known nonplatinum doublet.⁶⁷ Similarly, a randomized trial comparing gemcitabine-vinorelbine with carboplatin-vinorelbine showed similar ORR with more favorable median survival and better toxicity profile with the nonplatinum doublet.⁶⁸

Single-agent nonplatinum chemotherapeutic agents can also be considered in this population. The phase III CALGB

9730 trial⁶⁹ compared single-agent paclitaxel with platinum doublet of carboplatin and paclitaxel, and showed similar OS. Other nonplatinum agents that have shown efficacy and can be considered in this population include nab-paclitaxel,⁷⁰ gemcitabine,⁷¹ docetaxel,⁷² and pemetrexed.⁷³

Clinical interpretation. Patients with advanced NSCLC with contraindications to immunotherapy and platinum-based chemotherapy should be considered for alternative combination chemotherapy regimens such as gemcitabine with docetaxel,⁶⁷ gemcitabine with vinorelbine, or single-agent regimens such as paclitaxel, nab-paclitaxel, docetaxel, gemcitabine, or pemetrexed (nonsquamous histology), with decision guided by performance status, patient comorbidities, and patient goals.

Recommendation 3.4.

Literature review update. An initial phase II trial of carboplatin and paclitaxel with or without bevacizumab in advanced NSCLC demonstrated clinical efficacy of bevacizumab. Significant increase in fatal pulmonary hemorrhage was reported in patients with squamous cell histology (31%).⁷⁴ This led to the exclusion of squamous cell histology patients in the pivotal ECOG 4599 trial that led to the approval of bevacizumab.⁷⁵ In this trial, there was increased incidence of grade 3 to 5 neutropenia, fatigue, hypertension, venous embolism, and proteinuria in the bevacizumab group. A recent meta-analysis evaluating the safety and effectiveness of bevacizumab in the treatment of patients with nonsquamous NSCLC reported increased rate of hypertension (relative risk [RR], 5.34), hemorrhagic events (RR, 2.61), and proteinuria (RR, 7.55). There were no statistically significant differences in gastrointestinal perforation, arterial venous thrombotic events, or fatal events.⁷⁶

Clinical interpretation. Bevacizumab, when combined with chemotherapy, has been associated with increased risk of fatal pulmonary hemorrhage (especially in patients with squamous cell histology), increased rates of hypertension, proteinuria, and potential increased risk of arterial and venous thrombotic events. Bevacizumab has also not been studied in patients with ECOG PS > 1. Overall, the potential risks of bevacizumab likely outweigh the benefits and should be avoided for patients with squamous cell histology, clinically significant hemoptysis, significant cardiovascular disease, uncontrolled hypertension, and in patients with ECOG PS of >1.

Recommendation 3.5.

Literature review update. Addition of bevacizumab to platinum chemotherapy in treatment-naïve patients with advanced NSCLC followed by bevacizumab maintenance demonstrated significant survival benefit, especially in nonsquamous histology.⁷⁵

The AVAPERL study compared bevacizumab maintenance with or without pemetrexed after induction therapy with cisplatin, pemetrexed, and bevacizumab.⁷⁷ The PointBreak trial compared bevacizumab maintenance following carboplatin, paclitaxel, and bevacizumab induction with combined pemetrexed

and bevacizumab maintenance after induction carboplatin, pemetrexed, and bevacizumab. Both of these trials did not show an OS benefit, despite an improvement in PFS.⁷⁸

A pooled analysis of four RCTs (PointBreak, PARAMOUNT, PRONOUNCE, and JVBL) also failed to demonstrate an OS benefit for combination maintenance therapy with bevacizumab plus pemetrexed compared to pemetrexed maintenance alone.⁷⁹ Patients treated with bevacizumab plus pemetrexed combination maintenance also had higher rates of grade 3 to 5 AEs.

The phase III ECOG-ACRIN 5508 study compared maintenance therapy with either bevacizumab, pemetrexed, or bevacizumab combined with pemetrexed after induction carboplatin, paclitaxel, and bevacizumab. Combination maintenance therapy of bevacizumab plus pemetrexed failed to meet its primary end point of OS. There was also a significant increase in grade 3 to 4 toxicity in the combination arm compared to bevacizumab and pemetrexed (51% v 29% v 37%). The most common grade 3 to 4 toxicities were cytopenias and hypertension.⁸⁰

Clinical interpretation. Addition of bevacizumab maintenance therapy to pemetrexed maintenance after induction chemotherapy showed an improvement in PFS, which did not translate into an OS benefit. It is also associated with increased grade 3 to 4 toxicity compared to single-agent pemetrexed or bevacizumab. Given the lack of a survival benefit and increased toxicity, maintenance bevacizumab should not be routinely added to pemetrexed after induction chemotherapy.

SECOND-LINE AND SUBSEQUENT THERAPIES

Patients Previously Treated With Immune Checkpoint Therapy Without Chemotherapy

Recommendation 4.1

Literature review update. The data in support of platinum-doublet chemotherapy without checkpoint blockade in the subsequent line setting are primarily extrapolated from frontline studies conducted in the pre-immunotherapy era. The phase III ECOG 1594 study demonstrated a median OS of 7.9 months when evaluating multiple platinum doublet regimens including cisplatin-paclitaxel, cisplatin-gemcitabine, cisplatin-docetaxel, and carboplatin-paclitaxel, without significant differences in efficacy between regimens.⁸¹ In a phase III trial, the combination of cisplatin-pemetrexed was found to be noninferior to cisplatin-gemcitabine.⁶⁴ Subset analyses indicated an OS benefit for cisplatin-pemetrexed compared to cisplatin-gemcitabine in nonsquamous NSCLC (median OS 12.6 v 10.9 months; HR, 0.84), resulting in the adoption of pemetrexed-based platinum-doublet chemotherapy for this NSCLC histology.

Clinical interpretation. Although the evidence quality is low, there is strong consensus that histology-specific platinum-based doublet chemotherapy remains an effective option for patients who have no contraindications to, and have yet to receive, platinum doublet therapy. Data are limited regarding the role for continuation of anti-PD-(L)1 ICI in this setting.

Patients Previously Treated With Chemotherapy and Immune Checkpoint Therapy

Recommendation 4.2

Literature review update. The REVEL trial demonstrated an improvement in OS (10.5 v 9.1 months; HR, 0.86) and PFS (4.5 v 3.0 months; HR, 0.76) with docetaxel plus ramucirumab as compared to docetaxel alone.⁸² Grade ≥ 3 TRAEs were more commonly encountered in the combination group, including neutropenia (49% v 39%), fatigue (14% v 10%), and hypertension (6% v 2%).⁸² Real-world data confirm consistent benefit in the post-ICI setting, with median OS ranging from 11 to 21 months.⁸³⁻⁸⁷ In addition, as the control arm of several large phase III clinical trials in the subsequent-line setting in advanced NSCLC, docetaxel has demonstrated consistent clinical efficacy with a median PFS of 3.6-3.9 months and median OS of 9.8-11.8 months.^{88,89}

Clinical interpretation. Docetaxel, with or without ramucirumab, remains a standard subsequent-line therapy option after platinum-based chemotherapy and ICI therapy. The decision regarding whether or not to incorporate ramucirumab should be guided by several factors including disease histology and characteristics such as hemorrhage or invasion into adjacent structures, as well as patient comorbidities and tolerance of potential AEs.

Recommendation 4.3

Literature review update. In a randomized phase III noninferiority clinical trial, pemetrexed demonstrated equivalent OS compared to docetaxel (8.3 v 7.9 months; HR, 0.99) in chemotherapy pretreated advanced NSCLC, with fewer grade ≥ 3 hematologic toxicities.⁷³ In the phase III J-AXEL study, nab-paclitaxel was noninferior to docetaxel with a median OS of 16.2 versus 13.6 months (HR, 0.85 [95% CI, 0.68 to 1.07]), respectively, in advanced NSCLC previously treated with chemotherapy. Single-agent gemcitabine has also demonstrated modest clinical efficacy in small, published trials.⁹⁰

Clinical interpretation. Pemetrexed, nab-paclitaxel, and gemcitabine can be considered as subsequent-line therapy for patients, with selection guided by patient clinical status, disease histology, prior therapy, and toxicity profile.

Recommendation 4.4

Literature review update. In the two-stage, single-arm, phase II LUMINOSITY trial, 172 patients with advanced, pretreated, nonsquamous EGFR wild-type NSCLC with c-MET protein overexpression were enrolled and received the c-MET-targeted antibody-drug conjugate (ADC) teli-sotuzumab vedotin (Teliso-V) 1.9 mg/kg once every 2 weeks.²⁴ Among 78 efficacy-evaluable patients with high c-MET overexpression ($\geq 25\%$ TC with 3+ intensity), at a median follow-up of 20.2 months, the ORR was 34.6%, median PFS was 5.5 months, and median OS was 14.6 months. Among the 172 safety-evaluable population, common grade ≥ 3 TRAEs included peripheral sensory neuropathy (7.0%), increased ALT (3.5%), and pneumonitis (2.9%).

Clinical interpretation. Teliso-V may be offered to patients with advanced NSCLC with high c-MET overexpression who have received prior therapy, although evidence at this time is limited to a single-arm, noncomparative study. c-MET overexpression should be assessed in advanced NSCLC with disease progression on prior therapy. The ongoing global phase III TeliMET study is evaluating Teliso-V versus docetaxel in previously treated advanced EGFR wild-type NSCLC with high c-MET overexpression.

Recommendation 4.5

Literature review update. In the DESTINY-Lung01 trial, patients with advanced NSCLC with human epidermal growth factor receptor 2 (HER2) overexpression (immunohistochemistry [IHC] 2+ or 3+) received the HER2-targeted ADC trastuzumab deruxtecan (T-DXd). Among the 41 patient cohort treated at the now FDA-approved dose of 5.4 mg/kg once every 3 weeks, the ORR was 34.1%, with median PFS of 6.7 months (95% CI, 4.2 to 8.4) and median OS of 11.2 months (95% CI, 8.4 to not estimable [NE]).⁹¹ Common grade ≥ 3 treatment-emergent AEs included fatigue (7%), anemia (7%), and diarrhea (5%). The rate of all-grade drug-related interstitial lung disease was 5%. Among a 36-patient cohort of patients with HER2-overexpressing advanced NSCLC treated with T-DXd 5.4 mg/kg in the DESTINY-Lung03 phase Ib study, the reported ORR was 44.4%, with a median PFS of 8.2 months and a median OS of 17.1 months.⁹²

Clinical interpretation. While data in advanced NSCLC are limited to pooled subset data for patients with high HER2 overexpression (IHC 3+), T-DXd can be offered to patients with advanced HER2 IHC 3+ NSCLC who have experienced disease progression on chemotherapy and ICI. Evidence remains limited to phase II data, and further validation in randomized studies is needed.

Figures 1-3 provide the treatment algorithm based on these recommendations.

LIMITATION OF RESEARCH AND FUTURE RESEARCH

Tumor-Treating Fields

Multiple additional therapies are under active investigation in advanced NSCLC. A novel treatment approach, tumor-treating fields (TTFs), which utilizes electric fields to disrupt cancer growth processes, has demonstrated promising efficacy. In the phase III LUNAR trial, the addition of TTFs to standard investigator-choice subsequent therapy (ICI or docetaxel) resulted in an improved OS with the addition of TTFs. However, given this study was conducted in the setting of an evolving treatment landscape, the role for TTFs in the modern treatment paradigm is unknown. The ongoing phase III LUNAR-2 trial will help to address this.⁹³

VEGF-Based Therapies

The role and position of VEGF-based therapies in the treatment of advanced NSCLC is another important unanswered question. The VEGF-PD-1 bispecific antibody ivonescimab has shown promise compared to pembrolizumab in the phase III HARMONI-2 trial; however, data in a global population are needed to solidify these encouraging results. In addition, while the randomized phase II Lung-MAP S1800A showed promising results for the combination of pembrolizumab plus ramucirumab compared to investigator-choice chemotherapy in previously treated advanced NSCLC,⁹⁴ the phase III PRAGMATICA-LUNG study failed to show an OS benefit for pembrolizumab-ramucirumab compared to investigator-choice chemotherapy.⁹⁵

Subcutaneous and Intravenous Formulations

Two large phase III RCTs have been published investigating subcutaneous (SC) formulations of ICI in NSCLC. The 3475A-D77 trial involved a SC formulation of pembrolizumab, while IMscin001 Part 2 trial involved the use of SC atezolizumab. Both trials were large open-label RCTs (N = 377 and 371, respectively) designed to evaluate the pharmacodynamics, efficacy, and safety of a SC formulations. While both studies showed higher levels of serum drug concentration, there was no clear difference in toxicity.^{21,96} In the IMscin001 study, serious TRAEs were reported to be slightly higher in the intravenous arm (1.6% v 2.4%); however, there were two treatment-related deaths reported within the SC group.⁹⁶ Within the 3475A-D77, safety profiles were consistent between both groups, with no significant differences observed in overall AEs, grade 3 to 4 AEs, or treatment discontinuation rates.²¹ Similar to the IMscin001 study, there were numerically a higher number of treatment-related deaths in the intravenous arm (3.6% v 2.4%).²¹ A small percentage of injection site reactions were noted but these were reported to be minor in both studies.^{21,96}

Furthermore, there were no statistically significant differences in ORR ratio and median PFS in either study. In the

3475A-D77 trial, no differences were seen OS, while these data were immature in the IMscin001 trial.^{21,96} In a follow-up analysis of the 3475A-D77 trial, an analysis on time showed a 45.6% reduction in health care provider time. Overall, there was a 49.6% reduction in chair time (59.0 v 117.2 minutes).⁹⁷

Additional phase III trials involving the use of SC formulations of ICI have been conducted with nivolumab (CheckMate 67T) in renal cell carcinoma, as well as sasanlimab (CREST) in non-muscle-invasive bladder cancer. As in the previously described studies, CheckMate67T showed similar pharmacokinetics with no statistical difference in efficacy reported.⁹⁸ The CREST study differs in that it was not a pharmacokinetic study and a positive study investigating the addition of SC formulations of ICI to Bacillus Calmette-Guérin treatment in non-muscle-invasive bladder cancer.⁹⁹

With multiple phase III trials evaluating the pharmacodynamics, efficacy, and safety, SC formulations are emerging as a safe and effective alternative to intravenous formulations. While the benefit of reduced time toxicity is clear today, the financial landscape will complicate this equation after 2028 when major drugs such as pembrolizumab, nivolumab, and atezolizumab lose patent exclusivity.

To enhance the global relevance of this guideline, a summary of selected drug therapies that are currently available in specific countries outside of the United States along with their key supporting clinical trials is included in Appendix (Table A3). This table highlights several promising therapies that may be relevant to ongoing global clinical trial participation or offer insight into future treatment directions. This table will be updated on an annual basis to serve as an informational resource for clinicians worldwide.

PATIENT AND CLINICIAN COMMUNICATION

Treatment decisions for patients with NSCLC without a driver alteration are increasingly complex. For patients and their families, making treatment decisions can be overwhelming. Clear, honest, and compassionate communication is essential to ensure that treatment decisions are evidence-based and align with the patient's goals and preferences. See the Discussion Points Between Patients and Clinicians box.

For recommendations and strategies to optimize patient-clinician communication, see Patient-Clinician Communication: American Society of Clinical Oncology Consensus Guideline.¹⁰⁰

PATIENT ACCESS CONSIDERATIONS

Not all patients have consistent access to the resources needed to benefit from ASCO's expert recommendations on best practices for prevention, screening, palliative and

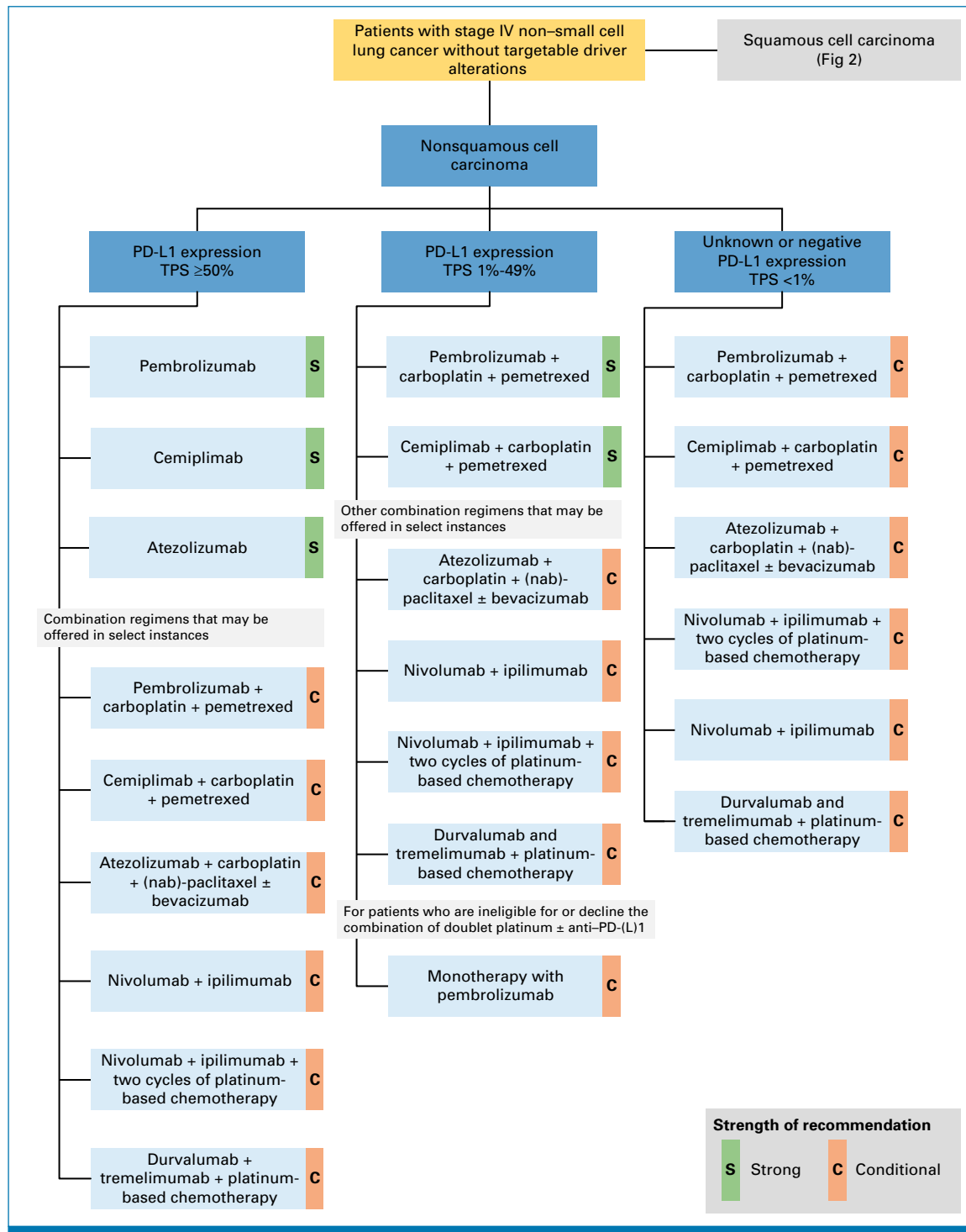


FIG 1. First-line treatment options for patients with stage IV non-small cell lung cancer without driver alterations and nonsquamous cell carcinoma. For recommendations with multiple treatment options of the same evidence quality and strength of recommendation, the decision of which agent to offer should be tailored based on discussion of efficacy and toxicity with each patient. nab, nanoparticle albumin-bound; NSCLC, non-small cell lung cancer; PS, performance status; TPS, tumor proportion score.

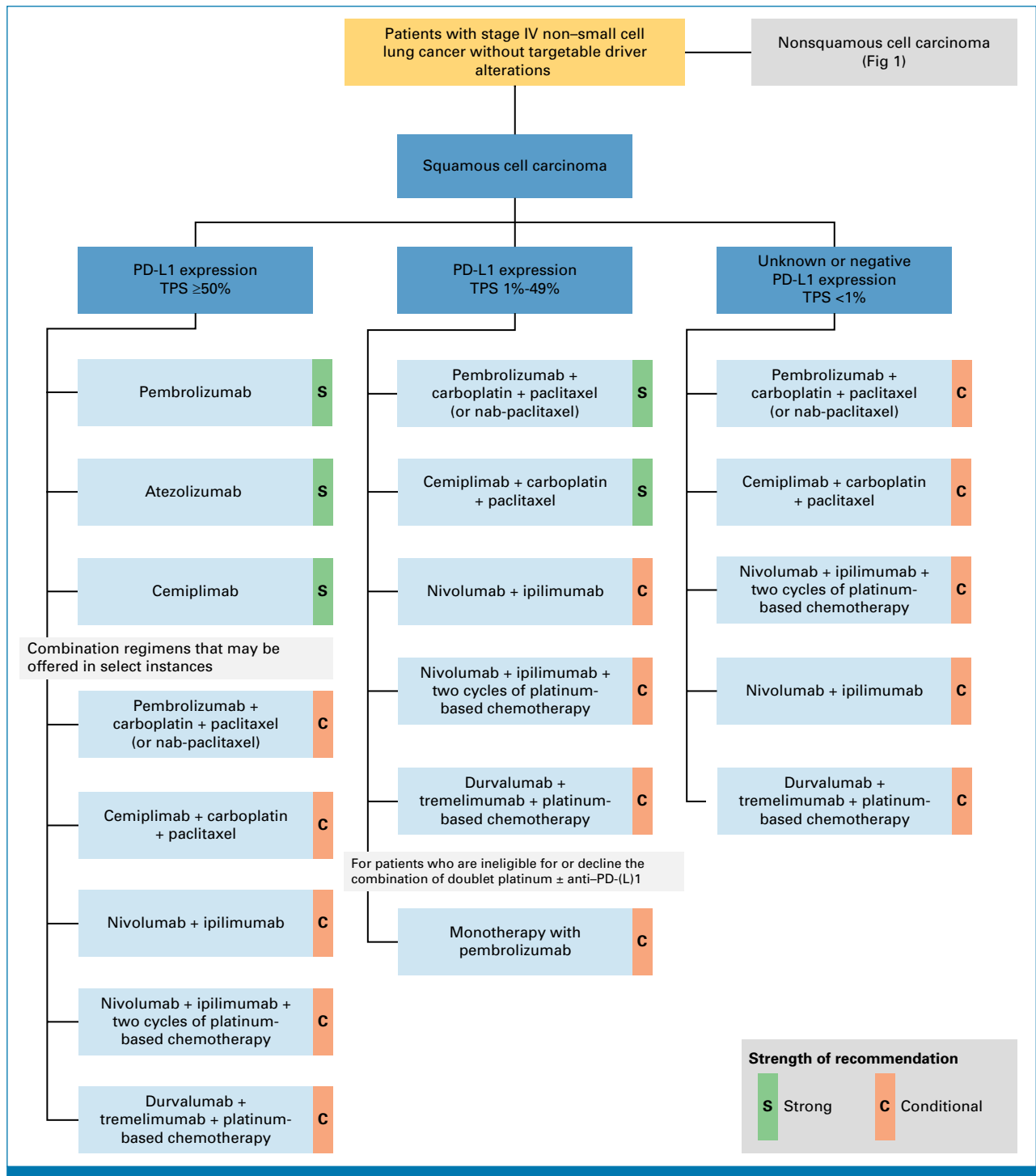


FIG 2. First-line treatment options for patients with stage IV non-small cell lung cancer without driver alterations and squamous cell carcinoma. For recommendations with multiple treatment options of the same evidence quality and strength of recommendation, the decision of which agent to offer should be tailored based on discussion of efficacy and toxicity with each patient. nab, nanoparticle albumin-bound; NSCLC, non-small cell lung cancer; PS, performance status; TPS, tumor proportion score.

supportive care, and disease management of cancer. In 2025, ASCO updated its guideline protocol template to include a patient access considerations section, allowing the Panel to consider patient access issues prior to starting the literature search.

In the United States, many patients remain unable to reap the benefits of innovative prevention and early detection programs, biomarker testing, and new cancer therapies due to challenges including lack of transportation, stable housing, adequate insurance coverage, food insecurity,

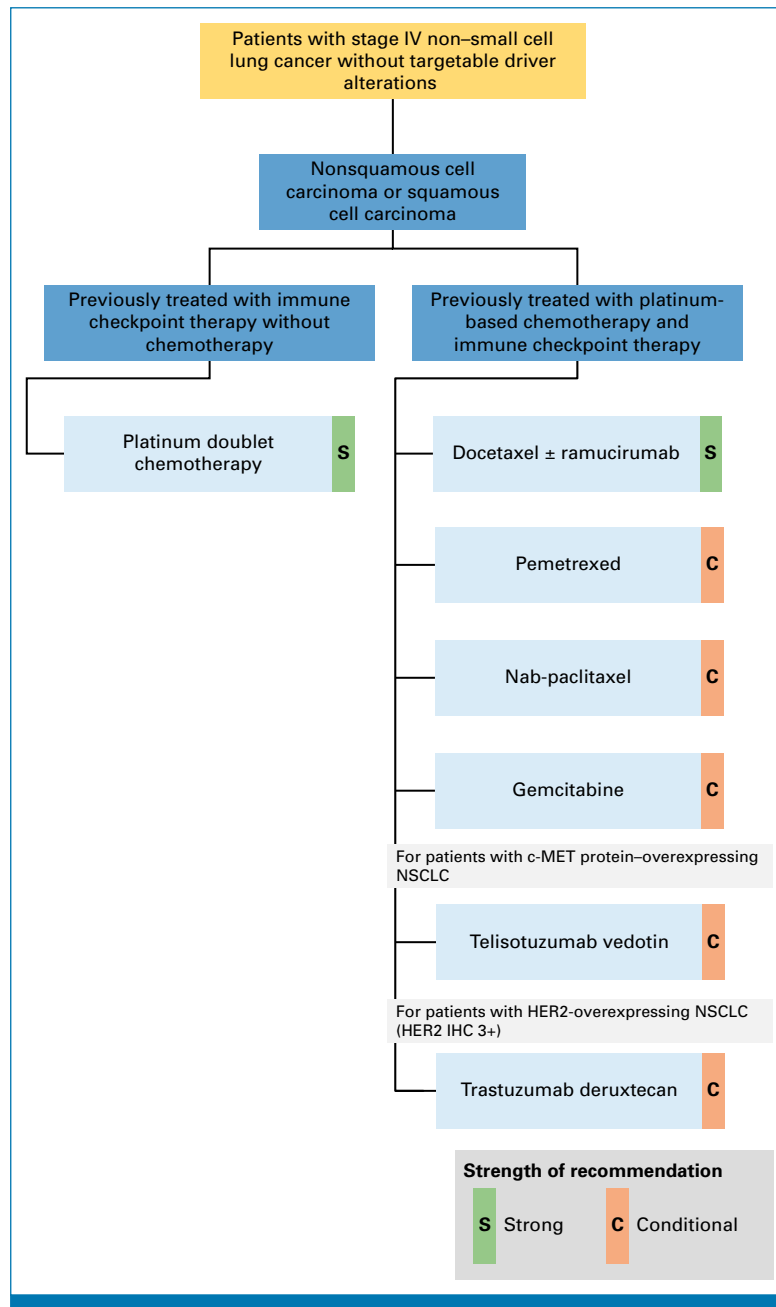


FIG 3. Second-line and subsequent treatment options for patients with stage IV non-small cell lung cancer without driver alterations. For recommendations with multiple treatment options of the same evidence quality and strength of recommendation, the decision of which agent to offer should be tailored based on discussion of efficacy and toxicity with each patient. HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; NSCLC, non-small cell lung cancer.

health literacy, proximity to a dedicated cancer center, and cost of treatment and other services.¹⁰¹ Additionally, some populations experience stigma along with barriers to cancer screening, prevention, and treatment that contribute to these cancer disparities.¹⁰² Black patients with metastatic NSCLC in the United States are significantly less likely than White patients to receive biomarker testing both before

first-line therapy (29.7% v 36.6%) and at any time during the disease course (43.8% v 54.7%). These disparities persisted after adjustment for baseline covariates. Social determinants of health also influence access to precision oncology; uninsured patients, those living in rural settings, and individuals with limited English proficiency are also less likely to undergo molecular testing.^{103,104}

DISCUSSION POINTS BETWEEN PATIENTS AND CLINICIANS

Goals of Care

Communicate treatment objectives (curative v palliative) using plain language centered around the patient's own goals and expectations.

Risks and Benefits of Therapeutic Options

Outline potential treatment options including a focus on treatment delivery, schedule, and AEs, incorporating patient-centered shared decision making.

Comorbidities

Incorporate medical comorbidities and patient-specific socioeconomic factors when determining a treatment regimen and making adjustments during treatment.

Biomarker and Genetic Testing

Explain all diagnostic testing results (imaging, molecular) and how they guide therapy, homing in on the personalized approach that is required to properly treat each patient.

Clinical Trials

Discuss the role of clinical trials and where they may (or may not) fit into the treatment course for a particular patient. Considerations include patient fitness for a study, line of therapy, and phase of trial (first-in human, phase II, phase III, etc)

Common Concerns, Including:

Treatment Burden and Symptom Management

Proactively address symptoms and quality of life, taking effort to make adjustments (eg, dose reductions, dose delays) where appropriate. Early integration of palliative care to maximize quality of life is strongly encouraged.

Psychosocial Concerns and Available Resources

Integrate patient psychosocial and cultural beliefs into care. Utilize resources at your disposal to connect patients with appropriate resources, such as patient advocacy groups (eg, LUNGEvity, Lung Cancer Research Foundation, etc), to educate and help identify a community.

Financial and Insurance Considerations and Resources

Incorporate financial considerations into decision making and utilize support networks to offset costs where possible (eg, transportation assistance, meal assistance, etc).

Survivorship Planning, Including Long-Term Follow-Up, and Monitoring

Where appropriate, integrate survivorship and long-term follow-up into routine care, including consideration of chronic side effects, symptom burden, and socioeconomic stresses.

Common Pitfalls to Avoid in Patient-Centered Communication

Information overload, overpromising or focusing on hope at the expense of reality, minimizing side effects and/or patient burdens, and making assumptions about patient goals and/or beliefs.

Further, geographic disparities can also impact the quality of care that patients receive. Rural patients are more likely to have worse survivorship outcomes and experience higher mortality rates compared to nonrural patients. This can be attributed, in part, to a lower density of specialists and dedicated cancer centers, as only 21% of nonmetropolitan counties in the United States have one or more practicing oncologists.¹⁰⁵ Additionally, patients may experience financial

toxicity from associated expenses: travel, time off work, or supportive care, which can limit adherence even after targeted therapy is prescribed.^{106,107} Programs that expand insurance coverage, negotiate drug pricing, or provide patient assistance are crucial in closing the existing gaps.¹⁰⁸ Opportunities to improve access to therapy in NSCLC lie at the intersection of policy, infrastructure, drug availability, and multidisciplinary collaboration.

MULTIPLE CHRONIC CONDITIONS

Creating evidence-based recommendations to inform treatment of patients with additional chronic conditions, a situation in which the patient may have two or more such conditions—referred to as multiple chronic conditions (MCC)—is challenging. Patients with MCC are a complex and heterogeneous population, making it difficult to account for all of the possible permutations to develop specific recommendations for care. In addition, the best available evidence for treating index conditions, such as cancer, is often from clinical trials whose study selection criteria may exclude these patients in order to avoid potential interaction effects or confounding of results associated with MCC. As a result, the reliability of outcome data from these studies may be limited, thereby creating constraints for expert groups to make recommendations for care in this heterogeneous patient population.

As many patients for whom guideline recommendations apply present with MCC, any treatment plan needs to consider the complexity and uncertainty created by the presence of MCC and highlight the importance of shared decision making regarding guideline use and implementation. Therefore, in consideration of recommended care for the target index condition, clinicians should review all other chronic conditions present in the patient and take those conditions into account when formulating the treatment and follow-up plan.

In light of these considerations, practice guidelines should provide information on how to apply the recommendations for patients with MCC, perhaps as a qualifying statement for recommended care. This may mean that some or all of the recommended care options are modified or not applied, as determined by best practice in consideration of any MCC.

COST IMPLICATIONS

Increasingly, individuals with cancer are required to pay a larger proportion of their treatment costs through deductibles and coinsurance.^{109,110} Higher patient out-of-pocket costs have been shown to be a barrier to initiating and adhering to recommended cancer treatments.^{111,112}

Discussion of cost can be an important part of shared decision-making.¹¹³ Clinicians should discuss with patients the use of less expensive alternatives when it is practical and feasible for treatment of the patient's disease and there are two or more treatment options that are comparable in terms of benefits and harms.¹¹³

GUIDELINE IMPLEMENTATION

ASCO guidelines are developed for implementation across health settings. Each ASCO guideline includes a community oncologist member on the panel. The additional role of this community oncologist member on the guideline panel is to

RELATED ASCO GUIDELINES

- Palliative Care for Patients With Cancer¹¹⁴ (<https://ascopubs.org/doi/10.1200/JCO.24.00542>)
- Patient-Clinician Communication¹⁰⁰ (<http://ascopubs.org/doi/10.1200/JCO.2017.75.2311>)
- Therapy for Stage IV Non–Small Cell Lung Cancer with Driver Alterations²⁶ (<https://ascopubs.org/doi/10.1200/JCO-25-02822>)
- Practical Assessment and Management of Vulnerabilities in Older Patients Receiving Systemic Cancer Therapy¹¹⁵ (<https://ascopubs.org/doi/10.1200/JCO.23.00933>)

assess the suitability of the recommendations for implementation in the community setting, but also to identify any other barrier to implementation a reader should be aware of. Barriers to implementation include the need to increase awareness of the guideline recommendations among front-line practitioners and survivors of cancer and caregivers, and also to provide adequate services in the face of limited resources. The guideline recommendations table and accompanying tools (available at www.asco.org/thoracic-cancer-guidelines) were designed to facilitate implementation of recommendations. This guideline will be distributed widely. ASCO guidelines are posted on the ASCO website and most often published in the *Journal of Clinical Oncology*.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

ADDITIONAL RESOURCES

For current information, including selected updates, supplements, and clinical tools and resources, visit www.asco.org/thoracic-cancer-guidelines. The Data Supplement (online only) for this guideline includes the search strategy and additional evidence tables. Guideline recommendations and algorithms are also available in the free ASCO Guidelines app (available for download in the [Apple App Store](#) and [Google Play Store](#)). Listen to key recommendations and insights from panel members on the [ASCO Guidelines podcast](#). The Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the methods used to develop this guideline. Patient information is available at www.cancer.org.

ASCO welcomes your comments on this guideline, including implementation challenges, new evidence, and how this guideline impacts you. To provide feedback, contact us at guidelines@asco.org. Comments may be incorporated into a future guideline update. To submit new evidence or suggest a

topic for guideline development, complete the form available at www.asco.org/guidelines.

INCLUSIVE LANGUAGE

ASCO is committed to promoting the health and well-being of all patients. ASCO guidelines are intended to apply to, and

be discussed clearly and compassionately with, all patients. For this reason, guideline authors use appropriately inclusive language. In instances in which the guideline draws upon data based on research in a specified population (eg, studies regarding women with ovarian cancer), the guideline authors describe the characteristics and results of the research as reported.

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EDITOR'S NOTE

This ASCO Clinical Practice Guideline provides recommendations, with comprehensive review and analyses of the relevant literature for each recommendation. Additional information, including a supplement with additional evidence tables, clinical tools and resources, and links to patient information at www.cancer.org, is available at www.asco.org/thoracic-cancer-guidelines.

EQUAL CONTRIBUTION

J.E.R., L.B., S.P., and N.B.L. were Expert Panel Co-Chairs.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Data analysis and interpretation: All authors

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Therapy for Stage IV Non–Small Cell Lung Cancer Without Driver Alterations: ASCO Living Guideline, Version 2026.3.0

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APPENDIX 2. GUIDELINE AND CONFLICTS OF INTEREST

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines ("Policy," found at www.asco.org/guideline-methodology). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting, or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

TABLE A1. Stage IV NSCLC Living Guideline Expert Panel Membership

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Logan Roof, MD	Ohio State University, Columbus, OH	Medical Oncology
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Yubao Wang, MD, PhD	John Muir Health, Walnut Creek, CA	Medical Oncology
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Nofisat Ismaila, MD	American Society of Clinical Oncology (ASCO), Alexandria, VA	ASCO Practice Guidelines Staff (Health Research Methods)

TABLE A2. Evidence Quality and Recommendation Rating Definitions

Term	Definition
Evidence quality	
High	We are very confident that the true effect lies close to that of the estimate of the effect
Moderate	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect
Very low	We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect
Strength of recommendation	
Strong	In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects All or almost all informed people would make the recommended choice for or against an intervention
Conditional	In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists Most informed people would choose the recommended course of action, but a substantial number would not

NOTE. GRADE Handbook.¹¹⁶**TABLE A3.** Lung Cancer Drugs Available Outside the United States

Drug (Generic)	Class/Mechanism	Indication	Country/Countries	Trial Name/Reference
Serplulimab	Anti-PD-1	First-line treatment	China	ASTRUM-004 (PMID: 38181795)
Toripalimab	Anti-PD-1	First-line treatment	China	CHOICE-01 (PMID: 36206498)
Nintedanib	VEGFR TKI	3rd/subsequent line	EU	NCT01948141
Camrelizumab	Anti-PD-1	First-line treatment	China	CameL (PMID: 39608979)
Ivonescimab	Anti-PD-1/VEGF	2nd-line PD-L1 –positive	China	HARMONi-A (PMID: 38820549) HARMONi-2 (PMID: 40057343)
S-1 (tegafur, gimeracil, and oteracil potassium)	Cytotoxic agent	NSCLC	Japan, Taiwan, China	EAST-LC PMC5834128
UFT (uracil and tegafur)	Cytotoxic agent	NSCLC	Japan	PMID: 15102997
Tislelizumab	Anti–PD-L1	First-line treatment for SCC and non-SCC NSCLC	China	RATIONALE-303 (NCT03358875) RATIONALE-307 (NCT03594747) RATIONALE-304 (NCT03663205)
Sugemalimab	Anti–PD-L1	First-line treatment for SCC and non-SCC NSCLC	China	GEMSTONE-302 (NCT03789604)

NOTE. The following therapies included are based on trials meeting the evidence requirement for inclusion in this guideline, and this list is for informational purposes only. These therapies are available in specific jurisdictions outside of the United States. Readers should consult local regulatory authorities/product monographs for the most up-to-date status.

Abbreviations: EU, European Union; NSCLC, non–small cell lung cancer; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor.